POINT DEFECTS AND HERMITIAN–YANG–MILLS FLOW ON SURFACES

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Abstract. We determine the leading metric and concentration scales of Hermitian–Yang–Mills flow for locally free extensions $0 \to L \to V \to \mathcal{I}_W M \to 0$ on a compact Kähler surface, where L, M are distinct line bundles of equal degree and W is a nonempty reduced finite scheme. Fix Hermitian–Einstein metrics on the line factors and their product as determinant. For every smooth initial metric with this determinant, the integrated squared norm $Z(T)$ of the fixed inclusion satisfies $Z(T) \sim v_0^2/(4\pi^2\sigma T)$, where v_0 is the volume and $\sigma = \sum_{p\in W} (\det Q_p)^{-1/2}$; Q_p is the quadratic form of the inclusion norm at p in orthonormal coordinates. On the radius $\{Z(T)/(v_0\sqrt{\det Q_p})\}^{1/2}$, the actual metrics converge smoothly to an explicit charge-one anti-self-dual metric. The inclusion half-height radius and the half-defect-energy radius have this same leading value. The proof corrects a compact family of metrics orthogonally to its small connection eigenspace and compares the resulting trajectory with every actual flow. It also gives a global inclusion profile and quantitative convergence outside shrinking balls. For arbitrary finite defects we prove $Z(T) = O(T^{-1})$; if a defect is nonreduced, then $TZ(T) \to 0$.

1. Introduction

Hermitian–Yang–Mills flow on a compact Kähler surface can converge away from finitely many points while its metric degenerates along a fixed holomorphic filtration. The limiting connection and point cycle identify the smooth graded bundle and the lost curvature charge. They do not, by themselves, determine the rate of the metric degeneration or the radius on which that charge concentrates. We determine both for a rank-two extension with distinct equal-degree line factors and reduced point defects.

Let (X, ω) be a connected compact Kähler surface, and suppose that a locally free rank-two bundle V fits into an exact sequence

$$0 \longrightarrow L \xrightarrow{\iota} V \longrightarrow \mathcal{I}_W M \longrightarrow 0. \tag{1}$$

The line bundles L, M have equal degree, and W is a nonempty finite scheme. Existence of the locally free middle bundle is a hypothesis. Fix HE metrics h_L, h_M, the induced determinant metric $h_L h_M$, and the inclusion ι. Write

$$\mathrm{d}V = \omega^2/2, \qquad v_0 = \int_X \mathrm{d}V, \qquad \kappa = 2\pi \deg_\omega L/v_0.$$

The metric flow is

$$H^{-1}\partial_T H = -2(i\Lambda_\omega F_H - \kappa I), \qquad \det H(0) = h_L h_M.$$

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Its inclusion norm, mass and full derivative energy are

$$u(T) = \iota^* H(T)/h_L, \qquad Z(T) = \int_X u(T)\,\mathrm{d}V, \qquad \mathcal{E}(T) = \int_X |D'_{H(T)\otimes h_L^{-1}}\iota|^2\,\mathrm{d}V. \tag{2}$$

In particular $Z' = -2\mathcal{E}$.

At a reduced zero p, use holomorphic coordinates orthonormal for the base metric at p. The norm of the inclusion in any smooth reference metric of the fixed determinant has quadratic term z^*Q_pz, with $Q_p > 0$. The determinant of this form is independent of that reference metric. Put

$$\lambda_p = \sqrt{\det Q_p}, \qquad \sigma = \sum_{p\in W} \lambda_p^{-1}, \qquad \alpha(T) = Z(T)/v_0, \qquad r_p(T) = \sqrt{\alpha(T)/\lambda_p}. \tag{3}$$

These constants retain the prescribed inclusion, line metrics, determinant and base metric.

Theorem 1.1. *Suppose $L \not\cong M$ and W is reduced. For every smooth initial metric of determinant h_Lh_M, the flow exists for all $T \geq 0$ and has the following properties.*

(1) *The actual inclusion mass and derivative energy satisfy*

$$Z(T) \sim \frac{v_0^2}{4\pi^2\sigma T}, \qquad cT^{-2} \leq \mathcal{E}(T) \leq CT^{-2} \tag{4}$$

for large T. The leading mass coefficient is independent of the initial metric.

(2) *There is a fixed smooth reference metric H_0 of the prescribed determinant, with $u_0 = \iota^*H_0/h_L$, and constants $C > 0$, $0 < \gamma < 1$, such that*

$$\sup_X \left| \frac{u(T)}{Z(T)} - \frac{1}{v_0}\frac{u_0}{u_0+\alpha(T)} \right| \leq CT^{-\gamma}. \tag{5}$$

For small $r > 0$ and large T, with constants independent of r, T,

$$\sup_{X\setminus B_r(W)} \left| \frac{u(T)}{Z(T)} - \frac{1}{v_0} \right| \leq C\{T^{-\gamma} + (Tr^2)^{-1}\}.$$

(3) *In fixed normalized holomorphic coordinates and frames near p, the metrics $H(T, r_p(T)y)$ converge smoothly on compact y-sets to*

$$H_{\mathrm{loc}}(y) = (1+|y|^2)I - \frac{|y|^2+2}{1+|y|^2}YY^*, \qquad Y = (y_1, -y_2)^{\mathsf{t}}. \tag{6}$$

This determinant-one metric is anti-self-dual and has curvature energy $8\pi^2$. More precisely, for every fixed $R > 0$,

$$\int_{B_{Rr_p(T)}(p)} |F_{H(T)}|^2\,\mathrm{d}V \longrightarrow 8\pi^2 \frac{R^4(R^2+3)}{(1+R^2)^3}.$$

The first inclusion half-height radius on each radial ray, defined by $u(T) = \alpha(T)/2$, and the first ball radius with curvature energy $4\pi^2$, are both $r_p(T)(1+o(1))$; the first equivalent is uniform in direction.

The local coordinates are Kähler normal at p. The frames normalize the line weights to one at p and put the inclusion in the form $\sqrt{\lambda_p}(z_1, -z_2)^{\mathsf{t}}$; Lemma 2.2 constructs them without changing the determinant. The theorem follows from Theorems 8.1 and 9.1. The proof also accounts for all defect charge in the ordered limits (109). The energy assertion in (4) is an order estimate, not a differentiated version of the mass equivalent.

For a general finite defect we obtain a separate one-sided statement, allowing repeated factors.

Theorem 1.2. *Under (1), without reducedness or distinctness of the line factors, every smooth fixed-determinant flow satisfies $Z(T) = O(T^{-1})$. If W has a nonreduced point, then*

$$\frac{-Z'(T)}{Z(T)^2} \longrightarrow +\infty, \qquad TZ(T) \longrightarrow 0.$$

This is Theorem 4.2. The nonreduced conclusion makes the restriction in Theorem 1.1 mathematically necessary for its positive mass coefficient. Repeated line factors present a different issue: the real small connection eigenspace has dimension three, whereas the scalar trajectory used here requires dimension one.

1.1. **Relation to earlier work.** Daskalopoulos–Wentworth identify the canonical connection limit of Yang–Mills flow on Kähler surfaces [DW04], with the correction [DW07c], and identify the blowup set and its multiplicities [DW07b]. Sibley–Wentworth extend the analytic–algebraic cycle comparison to compact Kähler manifolds [SW15]. We use their corrected multiplicity, which at an isolated surface point is the full length of the defect sheaf. These results determine the reflexive graded connection $D_{h_L} \oplus D_{h_M}$ and the point charges in our setting.

The predecessor [Wue26a] treats (1) with arbitrary nonempty finite W. It proves $Z = o(T^{-1/2})$ and fixed-compact convergence of u/Z to $1/v_0$, using a residue estimate for every splitting. Its remaining rate question asks whether $\log Z/\log T$ can still tend to $-1/2$, particularly for its toric tangent bundle. Theorem 1.1 gives the limit -1 in the reduced, distinct-factor class, including that example. Theorem 1.2 also strengthens the one-sided estimate throughout the predecessor's class.

Haiden–Katzarkov–Kontsevich–Pandit develop iterated-weight asymptotics for finite metric flows [HKKP23] and prove an actual bundle-flow theorem on compact Riemann surfaces [HKKP18, Theorem 1.1]. Their comparison method for integrably forced metric flows also applies in higher complex dimension [HKKP18, Section 4.2]. Lam proves bounded-remainder asymptotics on a fixed compact Kähler base when the graded object is locally free, under the stated rationality condition on the induced invariant inner product [Lam26, Theorem 1.1, Theorem 8.2]. The singular-graded extension is discussed in [Lam26, Remark 8.3], rather than included in that theorem. Our quotient $\mathcal{I}_W M$ is singular, and the coefficient and spatial profile depend on its local presentation.

The charge-one instanton is classical. We compute its holomorphic metric, inclusion norm and curvature in the normalization needed for the compact extension. For quantitative infinite-time bubbling on $\mathbb{R}^4$, Sire–Wei–Zheng construct Yang–Mills solutions with selected initial data and exponentially shrinking scale [SWZ26, Theorem 1]. Their base, data assertion and scale-selection problem differ from the fixed compact holomorphic extension considered here. The radius $r_p(T) \asymp T^{-1/2}$ is selected by the compact inclusion mass and its determinant data.

1.2. **The proof.** The starting point is an explicit smooth family H_a on the original compact bundle. It has inclusion norm $au_0/(u_0 + a)$, an exact full-derivative identity, and the local limit (6). A level-set residue estimate gives a lower bound for the actual dissipation, including the strict nonreduced conclusion. The compact family itself is not a solution of the metric flow.

The analytic step is to correct this family orthogonally to the full small eigenspace of the real operator $\frac{1}{2}D_a^*D_a$ on Hermitian trace-free endomorphisms. A fixed-base Sobolev estimate excludes additional modes concentrated at the points. Heat domination and the complementary gap give a logarithmic weighted inverse. The resulting nonlinear correction has uniformly small mean curvature, and parameter estimates identify its remaining one-dimensional curvature with its tangent up to an integrable error. The small eigenvalue has leading coefficient $2\pi^2\sigma/v_0$.

For an arbitrary initial metric, invariant metric distance first stays bounded by the forced comparison inequality. The small positive eigenvalue then gives polynomial decay of that distance, proving the mass coefficient and the global profile. Convexity of the Donaldson functional supplies the actual derivative-energy upper bound. Finally, rescaled elliptic and parabolic estimates upgrade the metric comparison to smooth inner convergence; Chern–Weil theory and the explicit local curvature integral identify the concentration radius and defect charge.

2. The extension and its normalization

Throughout the paper, (X,ω) is a connected compact Kähler surface and V is a holomorphic vector bundle in an exact sequence

$$0 \longrightarrow L \xrightarrow{\iota} V \xrightarrow{q} \mathcal{I}_W M \longrightarrow 0. \tag{7}$$

Here L and M are line bundles of equal ω-degree, W is a nonempty zero-dimensional analytic subscheme, and V has rank two and is locally free. Existence of this extension is part of the hypotheses. Write $\Sigma = \operatorname{Supp} W$ and $U = X \setminus \Sigma$, and also denote by q the composite $V \to \mathcal{I}_W M \hookrightarrow M$. The positive asymptotic theorem will require W to be reduced and $L \not\cong M$. We retain the general setting in this section and in the residue estimate.

2.1. **Metrics, norms and the fixed inclusion.** Use the determinant identification induced by (7). Choose Hermitian–Einstein metrics h_L, h_M and put

$$h_{\det} = h_L h_M, \qquad \mathrm{d}V = \frac{\omega^2}{2}, \qquad v_0 = \int_X \mathrm{d}V, \qquad \kappa = \frac{2\pi \deg_\omega L}{v_0}. \tag{8}$$

Degrees are $\int_X c_1(\,\cdot\,)\wedge\omega$. Both line curvature contractions equal κ. All integral norms use the volume form in (8). On Hermitian endomorphisms the real inner product is $\operatorname{tr}(st)$, and on skew-Hermitian endomorphisms it is $-\operatorname{tr}(st)$. Form norms are the ordinary real Kähler norms, extended to complex tensors. In Euclidean complex coordinates with $\omega_E = (i/2)\sum_j \mathrm{d}z_j\wedge\mathrm{d}\bar z_j$, one consequently has $|\mathrm{d}z_j|^2 = 2$.

For a smooth metric H on V with determinant $h_{\det}$, define

$$\begin{aligned} &\Phi_H = i\Lambda_\omega F_H - \kappa I, \quad u_H = \iota^* H/h_L, \\ &Z_H = \int_X u_H\,\mathrm{d}V, \qquad \mathcal{E}_H = \int_X |D'_{H\otimes h_L^{-1}}\iota|^2\,\mathrm{d}V. \end{aligned} \tag{9}$$

The endomorphism Φ_H is Hermitian and trace-free. The function u_H is smooth on all of X, vanishes exactly on Σ, and is positive on U. The inclusion in (7) is a fixed holomorphic section of $V\otimes L^{-1}$; it is not renormalized as time varies.

We use the nonnegative scalar Laplacian $\Delta_0 = \mathrm{d}^*\mathrm{d}$. Thus $-\Delta_0 = 4\sum_j \partial_{z_j}\partial_{\bar z_j}$ in Euclidean coordinates. The metric flow is

$$H^{-1}\partial_T H = -2\Phi_H, \qquad H(0) = H_{\mathrm{in}}, \qquad \det H_{\mathrm{in}} = h_{\det}. \tag{10}$$

The base, the holomorphic extension and the line metrics remain fixed along this equation.

2.2. **The qualitative flow input.** We state the standard convergence input in the form used below. Long-time existence holds for every smooth initial Hermitian metric on a compact Kähler base [Kob87, Theorem 6.8.25]. The determinant solves its scalar heat equation, so the initial condition in (10) preserves $h_{\det}$. The time in that reference is $t = 2T$.

The rank-one factors of (7) are stable of equal slope. Thus V is semistable, with graded sheaf $G = L \oplus \mathcal{I}_W M$ and reflexive hull $G^{**} = L \oplus M$. The corrected convergence theorem of Sibley–Wentworth [SW15, Theorem 1.1 and Remark 1.2, with the appended erratum] identifies the connection limit with

$$D_* = D_{h_L} \oplus D_{h_M}. \tag{11}$$

More precisely, after identifying the evolving metrics with a fixed Hermitian bundle, every sequence $T_j \to \infty$ has subsequences and unitary gauges giving smooth convergence on compact subsets of U to this connection. The singular set is Σ, and its multiplicity at p is $\dim_{\mathbb{C}} \mathcal{O}_{W,p}$. The corrected multiplicity counts the full local length, including nonreduced structure. We use this connection theorem and its point cycle; it supplies no rate on a shrinking neighbourhood.

The following scalar consequences were established in [Wue26a, Section 5]. We include their proof to fix the normalization and to separate the scalar assertion from a choice of limiting graded summand.

Proposition 2.1. *For every solution of (10),*

$$Z'(T) = -2\mathcal{E}(T), \qquad \sup_X u(T) \le CZ(T). \tag{12}$$

For every compact $K \subset U$,

$$\frac{u(T)}{Z(T)} \longrightarrow \frac{1}{v_0} \quad \text{in } C^\infty(K). \tag{13}$$

The constant and convergence times may depend on the initial metric.

Proof. The curvature contraction on $V \otimes L^{-1}$ is Φ_H. Differentiating the norm of the fixed section and using its Bochner identity [Kob87, Proposition 3.1.8] gives

$$\partial_T u = -2\langle \Phi_H \iota, \iota \rangle, \qquad -\Delta_0 u = 2|D'\iota|^2 - 2\langle \Phi_H \iota, \iota \rangle.$$

Integration on the compact surface proves the first identity in (12). No singular boundary is introduced: the section and its finite-time metric are smooth across Σ.

The mean-curvature heat identity, in the present time convention, is

$$(\partial_T + \Delta_0)|\Phi_H|^2 = -2|D_H \Phi_H|^2 \le 0; \tag{14}$$

see [Kob87, Proposition 6.9.1 and equation (6.9.2)]. Its maximum principle bounds $\|\Phi_H\|_\infty$ by the initial value. The scalar Bochner inequality therefore gives $-\Delta_0 u \ge -Cu$. The mean-value estimate on the fixed compact base yields $\sup u \le C \int u$, as asserted.

Transport the inclusion isometrically to the fixed Hermitian bundle used in the connection convergence theorem and divide it by $Z(T)^{1/2}$. The resulting holomorphic section j_T has integral squared norm one and a uniform pointwise norm bound. On a subsequence giving smooth connection convergence on U, interior estimates for its holomorphic equation give smooth subsequential convergence to a holomorphic section of $(L \oplus M) \otimes L^{-1}$ on U. Its coefficients extend over the isolated points by Hartogs' theorem. The resulting global bundle has zero contracted curvature, so its integrated Bochner identity makes the section parallel.

The pointwise bound excludes loss of section mass at the points: the mass in a union of radius-r coordinate balls is at most their volume times a fixed constant. First pass to the limit

on the complement and then let $r \downarrow 0$. The limiting parallel section still has squared integral norm one, and hence constant squared norm $1/v_0$. If the line factors coincide, different subsequences may give different parallel vectors; their squared norm is nevertheless the same. Since $|j_T|^2 = u/Z$ is unchanged by the isometries and gauges, a subsequence contradiction proves the full scalar convergence (13). □

2.3. **Local defect data.** In local holomorphic frames compatible with the determinant, the extension has the presentation

$$\iota = (f, -g)^{\mathsf{t}}, \qquad q = (g, f). \tag{15}$$

The ideal (f, g) has an isolated zero. In the regular local surface ring it is a regular sequence, so local complete intersection structure follows from the assumed locally free extension. Write $m_p = \dim_{\mathbb{C}} \mathcal{O}_{X,p}/(f, g)$.

At a reduced point choose holomorphic coordinates orthonormal for the base metric at p. For any smooth determinant-fixed reference metric H_0 the function $u_0 = \iota^* H_0/h_L$ has quadratic term

$$u_0(z) = z^* Q_p z + O(|z|^3), \qquad Q_p > 0.$$

The derivative of the inclusion is an isomorphism at a reduced zero. Its Gram determinant is independent of H_0 with fixed determinant: in the frames of (15),

$$d_p := \det Q_p = |\det \partial(f, g)(p)|^2 \frac{h_M(p)}{h_L(p)}, \qquad \lambda_p = \sqrt{d_p}, \qquad \sigma = \sum_{p \in \Sigma} \lambda_p^{-1}. \tag{16}$$

These expressions retain the fixed inclusion, base and line normalizations. Unit changes of orthonormal coordinates and compatible frames do not change d_p.

For later estimates we need more than a balanced value at the point: the reference metric must have a prescribed local form.

Lemma 2.2. *If W is reduced, there are Kähler normal holomorphic coordinates at each point, determinant-compatible holomorphic frames, and one smooth global metric H_0 with determinant $h_{\det}$ such that, on disjoint coordinate balls,*

$$\begin{gathered} \iota = \sqrt{\lambda_p}(z_1, -z_2)^{\mathsf{t}}, \qquad q = \sqrt{\lambda_p}(z_2, z_1), \\ H_0 = hI, \qquad h = \sqrt{h_L h_M}, \qquad u_0 = c_p(z)|z|^2, \qquad c_p = \lambda_p \sqrt{h_M/h_L}. \end{gathered} \tag{17}$$

The line weights equal one at the centre and have vanishing first derivatives. Consequently $h = 1 + O(|z|^2)$, $c_p = \lambda_p + O(|z|^2)$ and $\omega = \omega_E + O(|z|^2)$.

Proof. Choose Kähler normal coordinates and normalize the holomorphic line frames by a constant and the exponential of a linear function. This fixes their values and first metric jets. Complete their product to a determinant-compatible frame of V. Let $v(z)$ be the inclusion column and $J = \partial v(0)$. We first remove its higher terms by a determinant-one frame change while leaving the base coordinates fixed.

Write $v = A(z)z$, where $A(0) = J$ and A is holomorphic and invertible on a smaller ball. Put $B = JA^{-1}$ and $d = \det B$. Then $Bv = Jz$ and $d(0) = 1$. For $q_z = (-z_2, z_1)$ choose a holomorphic column w with $q_z w = d^{-1} - 1$; this is possible since the right side lies in the maximal ideal. The matrix $C = J(I + wq_z)J^{-1}$ fixes Jz and has determinant d^{-1}. Thus CB has determinant one and sends v to Jz.

A unitary change of the base-coordinate phase makes $\det J = -\lambda_p$. A constant determinant-one fibre change then makes $J = \sqrt{\lambda_p}\,\mathrm{diag}(1, -1)$. Determinant compatibility gives the quotient in (17). Prescribe $H_0 = \sqrt{h_L h_M}\, I$ on smaller balls. Its determinant

is exactly $h_{\det}$ and its inclusion norm is the one displayed. To extend these prescriptions, start with any global determinant-fixed metric. Express the local relative changes by trace-free self-adjoint logarithms, multiply by cutoffs supported in the disjoint balls, and exponentiate. The result agrees with the prescribed metrics near each point and retains the determinant everywhere. □

The hypotheses of this lemma are used only where reducedness is stated. A nonreduced ideal has a singular derivative and does not admit this linear local presentation.

3. The local metric and a compact comparison family

The local singularity fixes a useful smooth metric in its original holomorphic presentation. We first calculate that metric, including the normalization of its curvature charge, and then construct a family on the compact bundle. Its parameter is independent of flow time at this stage.

3.1. **An explicit metric at a reduced point.** On $\mathbb{C}^2$ use the Euclidean conventions of Section 2, the inclusion $z = (z_1, z_2)^{\mathsf{t}}$ and the quotient $(-z_2, z_1)$. A constant unitary sign change gives the presentation in (17). For $a > 0$ and $s = |z|^2$, put

$$w_a = \frac{a}{s+a}, \qquad P = \frac{zz^*}{s}, \tag{18}$$
$$H_a^{\mathrm{loc}} = w_a P + w_a^{-1}(I - P) = \frac{s+a}{a} I - \frac{s+2a}{a(s+a)} zz^*.$$

The final expression is defined also at the origin.

Lemma 3.1. *The metric (18) is smooth and positive on $\mathbb{C}^2$, equals I at zero, and has determinant one. Its Chern connection satisfies $i\Lambda_{\omega_E} F_a = 0$ and is anti-self-dual for the complex orientation. With the stated real tensor norms,*

$$u_a^{\mathrm{loc}} = \frac{as}{s+a}, \quad |D_a' z|^2 = \frac{4a^3}{(s+a)^3}, \quad |F_a|^2 = \frac{48a^2}{(s+a)^4}. \tag{19}$$

In particular the total curvature energy is $8\pi^2$. For a Euclidean radius-R ball,

$$\int_{B_R} |D_a' z|^2 \, \mathrm{d}V_E = 2\pi^2 a^2 \frac{R^4}{(R^2+a)^2}, \tag{20}$$

$$\int_{B_R} u_a^{\mathrm{loc}} \, \mathrm{d}V_E = \pi^2 a \left\{ \frac{R^4}{2} - aR^2 + a^2 \log(1 + R^2/a) \right\}, \tag{21}$$

$$\int_{B_R} |F_a|^2 \, \mathrm{d}V_E = 8\pi^2 \mathfrak{F}(R/\sqrt{a}), \qquad \mathfrak{F}(t) = \frac{t^4(t^2+3)}{(1+t^2)^3}. \tag{22}$$

Proof. The two projections in (18) have reciprocal positive eigenvalues. Its smooth matrix expression extends positivity and determinant one across zero. Use the holomorphic-frame convention $D' = \partial + H^{-1}\partial H$ and $F = \bar\partial(H^{-1}\partial H)$. The metric is equivariant under simultaneous unitary changes of base and fibre, so it suffices to calculate at $z = (r, 0)$.

For a general radial $w > 0$, set $f = \log w$ and $F_{\bar j k} = \partial_{\bar z_j}(H^{-1}\partial_{z_k} H)$. Differentiating the projection expression gives

$$F_{\bar 1 1} = (f' + s f'') \operatorname{diag}(1, -1), \qquad F_{\bar 2 2} = \left(f' + \frac{1 - w^2}{s} \right) \operatorname{diag}(1, -1).$$

For example, $H^{-1}\partial_{z_2}H$ has lower-left entry $(w^2-1)/r$ at the axis. When taking the next derivative, the derivative of H^{-1} contributes $-(w-w^{-1})^2/s$ to the first diagonal entry. Including this term gives the displayed second formula. The contraction therefore vanishes exactly when

$$s^2 f'' + 2sf' + 1 - e^{2f} = 0.$$

For $f = \log a - \log(s+a)$, the numerator after multiplication by $(s+a)^2$ is $s^2 - 2s(s+a) + (s+a)^2 - a^2 = 0$. The curvature is trace-free and of type $(1,1)$, so its primitivity is equivalent to anti-self-duality.

The inclusion lies in the range of P, which gives its norm. At the axis its covariant derivative has coefficient columns $(w_a, 0)^{\mathsf{t}}$ and $(0, w_a^2)^{\mathsf{t}}$. Each has squared H_a^{loc}-norm w_a^3. Since $|\mathrm{d}z_j|^2 = 2$, the section-energy density is $4w_a^3$. For a radial scalar function $U(s)$, $-\Delta_0 U = 4sU'' + 8U'$. Substitution also verifies $-\Delta_0 u_a^{\mathrm{loc}} = 2|D'_a z|^2$.

For completeness, put $c = a/(s+a)^2$. The four curvature coefficients in the holomorphic frame at the axis are

$$F_{\bar{1}1} = -c\,\mathrm{diag}(1,-1), \quad F_{\bar{2}2} = c\,\mathrm{diag}(1,-1),$$
$$F_{\bar{1}2} = -\frac{2a^2}{(s+a)^3}E_{21}, \quad F_{\bar{2}1} = -\frac{2}{s+a}E_{12}.$$

Conjugating by $H_a^{1/2}$ to a unitary frame makes the two off-diagonal coefficients both $-2c$. The four forms $\mathrm{d}\bar{z}_j \wedge \mathrm{d}z_k$ are mutually orthogonal and have squared norm four. The sum of the squared matrix norms is $12c^2$, giving $|F_a|^2 = 48c^2$.

The radial Euclidean measure is $\pi^2 s\,\mathrm{d}s$. Integrating the three densities gives (20)–(22). In particular

$$\int_{\mathbb{C}^2} |F_a|^2\,\mathrm{d}V_E = 48\pi^2 \int_0^\infty \frac{t}{(1+t)^4}\,\mathrm{d}t = 8\pi^2.$$

With our skew-Hermitian trace convention, anti-self-duality gives $\mathrm{tr}(F_a \wedge F_a) = |F_a|^2 \mathrm{d}V_E$, fixing the positive charge sign. □

The normalized inclusion is $s/(s+a)$. Its half-height radius is $\sqrt{a}$, and (22) has half its total energy at the same radius because $\mathfrak{F}(1) = 1/2$. Under $z = \sqrt{a}\,y$ the curvature measure becomes exactly $48(1+|y|^2)^{-4}\mathrm{d}V_y$. These are identities for a stationary local metric. A nonconstant choice $a = a(T)$ gives a nonzero metric velocity while the curvature contraction remains zero, so time dependence requires a separate compact correction.

3.2. **A family on the original compact bundle.** Return to the general extension (7) and fix any smooth H_0 with determinant $h_{\det}$. Using h_L for the adjoint of the inclusion, put $S = \iota\iota^\dagger$ and $u_0 = \mathrm{tr}\,S = \iota^* H_0/h_L$. Then $S^2 = u_0 S$, and both S and u_0 vanish on Σ. Define

$$T_a = \frac{u_0+a}{a} I - \frac{u_0+2a}{a(u_0+a)} S, \qquad H_a = H_0 T_a, \qquad a > 0. \tag{23}$$

This formula is intrinsic on V. The product $H_0 T_a$ means the Hermitian form with that matrix in a local frame.

Proposition 3.2. *The metrics (23) are smooth, positive and have determinant $h_{\det}$. If $e_a = |D'_{H_a \otimes h_L^{-1}}\iota|^2$, then*

$$u_a = \frac{a u_0}{u_0+a}, \qquad e_a = \left(\frac{a}{u_0+a}\right)^3 e_0, \qquad Z_a = a v_0 - a^2 \int_X \frac{\mathrm{d}V}{u_0+a}. \tag{24}$$

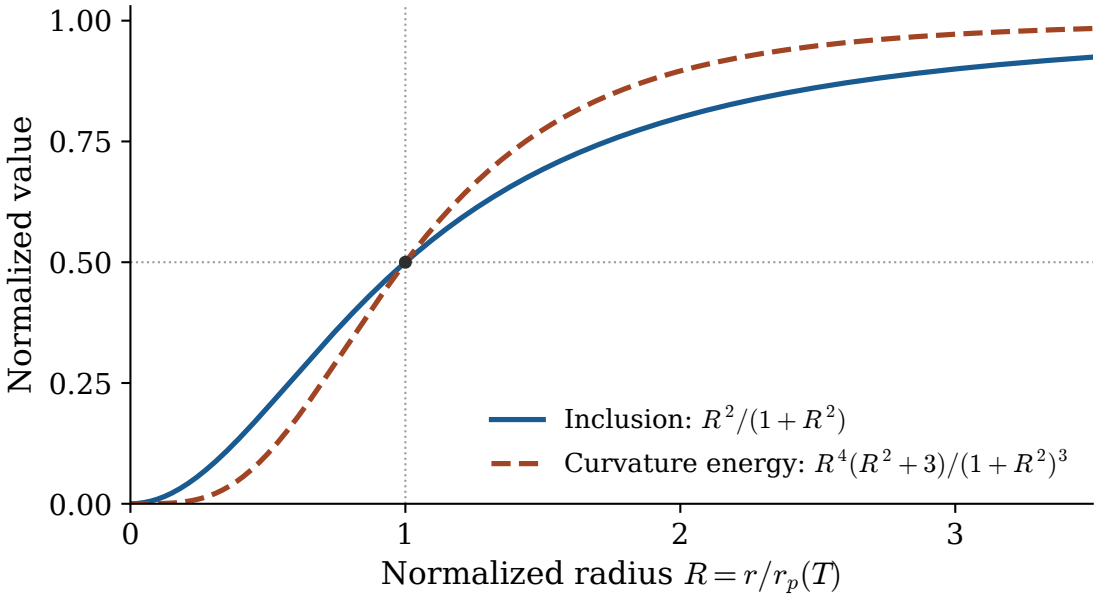


FIGURE 1. The exact local inclusion profile and the fraction of the charge-one curvature energy inside a ball. Each takes the value $1/2$ at normalized radius one. The curves depict the explicit local metric; their identification as actual-flow limits is proved in Sections 8 and 9.

For every finite W, $Z_a/a \to v_0$ and $u_a/Z_a \to 1/v_0$ smoothly on compact subsets of U. If W is reduced, then

$$\mathcal{E}_a = C(H_0)a^2 + o(a^2), \qquad C(H_0) = \pi^2 \sum_{p\in\Sigma} \frac{\operatorname{tr} Q_p}{\det Q_p}, \qquad Z_a = av_0 - a^2 I_0 + o(a^2), \tag{25}$$

where $I_0 = \int_X u_0^{-1}\mathrm{d}V < \infty$. Furthermore $C(H_0) \geq 2\pi^2\sigma$, with equality for the reference metric of Lemma 2.2.

Proof. On U, $P = S/u_0$ is the H_0-orthogonal projection onto the inclusion line, and

$$T_a = \frac{a}{u_0+a}P + \frac{u_0+a}{a}(I-P).$$

Its eigenvalues are positive reciprocals. The global expression (23) is smooth at Σ, where $T_a = I$. This proves the metric and determinant assertions. Evaluation on the inclusion gives the first formula in (24), and integration gives the last.

To calculate the full derivative, let τ be the H_0-orthogonal right inverse of q on U and write $\bar\partial\tau = \iota\beta$. The same splitting is orthogonal for H_a; its line and quotient metrics are $(u_a h_L, u_a^{-1}h_M)$. For any metric diagonal in this splitting with induced inclusion norm u, the tangential and normal Chern derivatives are orthogonal, and their squared norms are respectively

$$e = \frac{|\partial u|^2}{u} + u^3|\beta|_0^2. \tag{26}$$

Here $|\beta|_0$ uses the fixed metric $h_L h_M^{-1}$ and the base metric. Indeed the tangential component is $(\partial \log u)\iota$, while the adjoint of the off-diagonal Dolbeault block has the two quotient weights

and the additional inclusion weight. Since $\partial u_a = a^2 \partial u_0/(u_0+a)^2$, both terms in (26) acquire the factor $a^3/(u_0+a)^3$. This proves the middle identity on U, and smoothness extends it across Σ.

The bound $0 \le u_a/a \le 1$ and dominated convergence give $Z_a/a \to v_0$. On each compact subset of U, the rational formula converges with every fixed derivative.

At a reduced zero, $e_0(p) = 2\operatorname{tr} Q_p$ because the connection terms multiply the vanishing inclusion and $|\mathrm{d}z_j|^2 = 2$. On a small ball $u_0 \asymp |z|^2$ and e_0 is bounded. Substitute $z = \sqrt{a}\, y$ in the energy identity and divide its integral by a^2. The integrand, extended by zero outside the rescaled ball, is bounded by $C(1+c|y|^2)^{-3}$, integrable on $\mathbb{R}^4$. It converges to $e_0(p)(1+y^* Q_p y)^{-3}$. Thus its integral tends to

$$\frac{e_0(p)}{\det Q_p} \int_{\mathbb{C}^2} (1+|y|^2)^{-3}\, \mathrm{d}V_y = \pi^2 \frac{\operatorname{tr} Q_p}{\det Q_p}.$$

On the fixed exterior the energy is $O(a^3)$. Summing proves the energy assertion. Also u_0^{-1} is locally bounded by $C|z|^{-2}$, which is integrable in four real dimensions. Dominated convergence in the last formula of (24) gives the mass expansion.

Finally $\det Q_p = d_p$ is fixed, and $\operatorname{tr} Q_p \ge 2\sqrt{d_p}$, with equality exactly when $Q_p = \sqrt{d_p}\, I$. The normalized reference metric has this value at every point, proving the last assertion. □

3.3. **Matching the local and exterior descriptions.** From now on, when W is reduced we use the one normalized reference metric in Lemma 2.2. On its local ball, put $s = |z|^2$, $c = c_p(z)$ and $h = \sqrt{h_L h_M}$. The compact metric is

$$H_a = h\{wP + w^{-1}(I-P)\}, \qquad w = \frac{a}{a+cs}, \qquad u_a = \frac{acs}{a+cs}, \tag{27}$$

where P projects onto $(z_1, -z_2)^{\mathsf{t}}$. At the scale $z = \sqrt{a/\lambda_p}\, y$, its smooth matrix expression converges in every fixed number of derivatives on compact y-sets to

$$H_{\mathrm{loc}}(y) = (1+|y|^2)I - \frac{|y|^2+2}{1+|y|^2} YY^*, \qquad Y = (y_1, -y_2)^{\mathsf{t}}. \tag{28}$$

Indeed $u_0/a \to |y|^2$, $H_0 \to I$, and $S/a \to YY^*$ smoothly there; all denominators are bounded below. The rescaled base metric tends to the Euclidean metric. This is the local metric of Lemma 3.1 at parameter one, with its fixed unitary sign change.

On the compact exterior $u_a/a \to 1$. The exact common formula (23) joins these descriptions without changing the holomorphic bundle or its inclusion. In particular

$$Z_a = av_0 + O(a^2), \qquad \mathcal{E}_a = 2\pi^2 \sigma a^2 + o(a^2). \tag{29}$$

These are static identities and asymptotics. The next sections estimate the mean curvature, correct its orthogonal component and prove the parameter control needed to compare the resulting path with actual flow.

4. A level-set residue estimate for dissipation

The local residue controls the full inclusion derivative energy on sublevel sets of its norm. This gives an actual-flow bound before constructing a comparison trajectory. The estimate allows nonreduced defects and coincident line factors.

Fix a coordinate ball B_R about one $p \in \Sigma$, with no other defect in its closure, and use (15). The base and line metrics extend smoothly to a neighbourhood of the closed ball. Set

$$K_p(R) = \sup_{\overline{B}_R} |\mathrm{d}f \wedge \mathrm{d}g|_\omega \sqrt{h_M/h_L}, \qquad m_p = \dim_{\mathbb{C}} \mathcal{O}_{X,p}/(f,g). \tag{30}$$

The finite map germ (f, g) has a Jacobian which is not identically zero, so $K_p(R) > 0$ for small $R > 0$. No evolving bundle metric enters this constant.

Lemma 4.1. *For every smooth Hermitian metric H on $V|_{B_R}$ with the prescribed determinant, let $u = \iota^* H/h_L$ and $e = |D'_{H\otimes h_L^{-1}}\iota|^2$. Then*

$$\int_{B_R} e \,\mathrm{d}V \geq \frac{4\pi^2 m_p}{K_p(R)} \left(\min_{\partial B_R} u\right)^2. \tag{31}$$

Proof. Let τ be the H-orthogonal right inverse of q on the punctured ball, with $\bar\partial\tau = \iota\beta$. The form β is $\bar\partial$-closed, so $\Omega = \mathrm{d}f \wedge \mathrm{d}g \wedge \beta$ is closed. Its integral over every small outward sphere is

$$\int_{S_r} \Omega = 4\pi^2 m_p. \tag{32}$$

Here is the normalization and its independence of the splitting. The right inverse $\tau_0 = (\bar g, \bar f)^{\mathsf t}/(|f|^2 + |g|^2)$ has extension form

$$\beta_0 = \frac{\bar f\,\mathrm{d}\bar g - \bar g\,\mathrm{d}\bar f}{(|f|^2 + |g|^2)^2}.$$

The Bochner–Martinelli degree formula gives (32) for β_0, with local algebra length m_p; see [Suwa05, Chapter II, Section 5, Example 5.5; Chapter IV, Section 2, Theorems 2.3–2.4]. Any other right inverse differs by ιa, and its extension form differs by $\bar\partial a$. The change in Ω is the exact form $\mathrm{d}(a\,\mathrm{d}f \wedge \mathrm{d}g)$, whose sphere integral vanishes.

The orthogonal splitting has induced metrics $(uh_L, u^{-1}h_M)$. The same Chern calculation as in (26) gives

$$e = \frac{|\partial u|^2}{u} + u^3|\beta|_0^2.$$

The $(1,0)$ part of $\mathrm{d}u$ wedges to zero with $\mathrm{d}f \wedge \mathrm{d}g$. The product inequality for the remaining forms, followed by $2AB \leq A^2 + B^2$, yields

$$\begin{aligned} |\mathrm{d}(u^2\Omega)| &\leq 2u|\partial u|\,|\mathrm{d}f \wedge \mathrm{d}g|_\omega \sqrt{h_M/h_L}\,|\beta|_0 \\ &\leq K_p(R)\left(\frac{|\partial u|^2}{u} + u^3|\beta|_0^2\right). \end{aligned} \tag{33}$$

For top-degree complex forms the norm is the modulus of their coefficient relative to $\mathrm{d}V$. In particular $|\mathrm{d}z_1 \wedge \mathrm{d}z_2| = 2$ in the Euclidean convention; this factor is retained in $K_p(R)$.

Put $b_R = \min_{\partial B_R} u > 0$. For a regular value $0 < b < b_R$, the set $U_b = \{x \in B_R : u(x) < b\}$ has compact closure in B_R and smooth boundary, on which $u = b$. It contains p. Stokes' theorem for Ω gives $\int_{\partial U_b} \Omega = 4\pi^2 m_p$; any component not containing p contributes zero.

We justify the puncture removal also for $u^2\Omega$. In fixed smooth coefficient norms put $|F| = (|f|^2 + |g|^2)^{1/2}$. The orthogonal inverse is

$$\tau = \frac{H^{-1}q^*}{qH^{-1}q^*}.$$

Smooth positivity gives $qH^{-1}q^* \asymp |F|^2$ and $u \asymp |F|^2$. Differentiating, with constants allowed to depend on this fixed metric, gives

$$|\bar\partial\tau| \leq C\left(\frac{|\mathrm{d}F|}{|F|^2} + \frac{1}{|F|}\right), \qquad |\beta| \leq C\left(\frac{|\mathrm{d}F|}{|F|^3} + \frac{1}{|F|^2}\right).$$

Consequently $|u^2\beta| \le C(|F|\,|\mathrm{d}F| + |F|^2)$ is bounded. Since $\mathrm{d}f \wedge \mathrm{d}g$ is bounded, the integral of $u^2\Omega$ over a radius-ε sphere tends to zero. Also (33) makes its exterior derivative integrable, since e is smooth on the whole ball. Stokes' theorem on $U_b \setminus B_\varepsilon$ followed by $\varepsilon \downarrow 0$ thus gives

$$4\pi^2 m_p b^2 = \left| \int_{U_b} \mathrm{d}(u^2\Omega) \right| \le K_p(R) \int_{U_b} e \,\mathrm{d}V.$$

Choose regular values tending upward to b_R to obtain (31). The metric-dependent bounds above justify puncture removal only; the final constant is the fixed geometric constant (30). □

Theorem 4.2. *Under the general hypotheses (7), every solution of (10) satisfies*

$$Z(T) = O(T^{-1}). \tag{34}$$

If W has a nonreduced point, then

$$\frac{-Z'(T)}{Z(T)^2} \longrightarrow +\infty, \qquad TZ(T) \longrightarrow 0. \tag{35}$$

If every point is reduced, define

$$C_* = \frac{4\pi^2\sigma}{v_0^2}. \tag{36}$$

Then

$$\liminf_{T\to\infty} \frac{-Z'(T)}{Z(T)^2} \ge C_*, \qquad \limsup_{T\to\infty} TZ(T) \le C_*^{-1}. \tag{37}$$

Proof. Take disjoint fixed coordinate balls about the defects. The actual metric is smooth at every finite time with the fixed determinant, so Lemma 4.1 applies on each ball. By Proposition 2.1, $\min_{\partial B_R(p)} u(T)/Z(T) \to 1/v_0$. The exact dissipation identity therefore gives

$$\liminf_{T\to\infty} \frac{-Z'(T)}{Z(T)^2} \ge \frac{8\pi^2}{v_0^2} \sum_{p\in\Sigma} \frac{m_p}{K_p(R)}. \tag{38}$$

For any such fixed balls the right side is positive. The derivative of $1/Z$ is thus eventually bounded below by a positive constant, which proves (34).

At a reduced zero, (16) and the real two-form norm give

$$|\mathrm{d}f \wedge \mathrm{d}g|_\omega(p) = 2|\det \partial(f,g)(p)|, \qquad K_p(R) \longrightarrow 2\sqrt{d_p}.$$

Let $R \downarrow 0$ after taking the time limit in (38). Since $m_p = 1$, this proves the first assertion in (37). For every $c < C_*$ one has $(1/Z)' \ge c$ for all sufficiently large times. Integrate and then let $c \uparrow C_*$ to prove the second assertion.

At a nonreduced zero the derivative of (f,g) cannot be invertible: the holomorphic inverse theorem would otherwise give the maximal ideal and length one. Hence $\mathrm{d}f \wedge \mathrm{d}g$ vanishes at that point and $K_p(R) \to 0$. Its summand in (38) becomes arbitrarily large. Given any positive constant, first choose a fixed radius small enough and then take time sufficiently large. This proves $(1/Z)' \to +\infty$. Integration gives $TZ(T) \to 0$. □

For reduced points the static metrics (29) satisfy $2\mathcal{E}_a/Z_a^2 \to C_*$. Thus the coefficient in (37) is attained by an explicit family of smooth compact metrics. This does not yet establish its attainment along the actual flow. The remainder of the paper supplies the elliptic correction and dynamical comparison in the distinct-factor case. The nonreduced conclusion (35) is an upper-rate statement and does not select a sharp power or a logarithmic correction.

5. Small connection eigenvalues and mean curvature

We first identify all small eigenvalues of a real connection operator for the compact metrics H_a. The argument applies to an arbitrary finite defect. We then use the normalized reference metric of Lemma 2.2 to estimate the mean curvature when the defect is reduced. All constants in this section concern the fixed extension, base metric and reference metric.

5.1. **The real connection operator.** Let D_a be the Chern connection of H_a. On the real bundle of H_a-Hermitian trace-free endomorphisms, set

$$\mathfrak{q}_a(s,t) = \operatorname{Re}\int_X \langle D_a'' s, D_a'' t\rangle \,\mathrm{d}V = \frac{1}{2}\operatorname{Re}\int_X \langle D_a s, D_a t\rangle \,\mathrm{d}V, \qquad \|s\|_a^2 = \int_X \operatorname{tr}(s^2)\,\mathrm{d}V. \tag{39}$$

The second equality follows from Hermitian adjunction and real polarization. The associated self-adjoint operator is $\mathcal{L}_a = \frac{1}{2}D_a^* D_a$. We write its eigenvalues, counted with multiplicity, as $\lambda_1(a) \le \lambda_2(a) \le \cdots$. At a metric with noncentral mean curvature the complex operator $(D_a'')^* D_a''$ on all endomorphisms need not preserve Hermitian sections. The real operator in (39) is the one used throughout the proof.

Let $\mathcal{P}$ be the space of parallel Hermitian trace-free endomorphisms of the smooth graded connection D_* in (11). Then

$$k = \dim_{\mathbb{R}} \mathcal{P} = \begin{cases} 1, & L \not\simeq M, \\ 3, & L \simeq M. \end{cases} \tag{40}$$

Indeed a nonzero parallel off-diagonal map is a line-bundle isomorphism. If the two line bundles are isomorphic, their HE metrics differ, after identification, by a positive constant; the logarithmic ratio is harmonic. A normalized isomorphism is therefore parallel. It supplies two real off-diagonal directions in addition to the diagonal trace-free direction.

Proposition 5.1. *For the family (23) with any smooth reference metric of the fixed determinant, all $\lambda_j(a)$ are positive. There are $a_0, c > 0$ such that*

$$\lambda_k(a) \longrightarrow 0, \qquad \lambda_{k+1}(a) \ge c \quad (0 < a \le a_0). \tag{41}$$

If W is reduced, then $\lambda_k(a) \le Ca$. The small spectral space converges, on every compact subset of $X \setminus \operatorname{Supp} W$, to $\mathcal{P}$ under the unitary identifications specified below. In particular, it contains every normalized mode whose connection energy tends to zero.

Proof. Put $U = X \setminus \Sigma$, where $\Sigma = \operatorname{Supp} W$, and let τ be the H_0-orthogonal right inverse of the quotient map. The splitting (ι, τ) has metric $\operatorname{diag}(u_a h_L, u_a^{-1} h_M)$. Consequently

$$F_a(l,m) = u_a^{-1/2}\iota(l) + u_a^{1/2}\tau(m) \tag{42}$$

is a unitary isomorphism from $(L \oplus M, h_L \oplus h_M)$ to (V, H_a) over U. Write $\bar\partial\tau = \iota\beta$, with the fixed line metric on $\operatorname{Hom}(M,L)$ understood. The diagonal connection differences in this unitary splitting are opposite copies of $\frac{1}{2}(\partial - \bar\partial)\log u_a$; the off-diagonal blocks have norm $u_a|\beta|_0$. Thus

$$|F_a^* D_a - D_*| \le C\big(|\mathrm{d}\log u_a| + u_a|\beta|_0\big). \tag{43}$$

Since $\log u_a = \log a - \log(1 + a/u_0)$, this difference and all its fixed derivatives are $O_K(a)$ on each $K \Subset U$.

The scalar Sobolev inequality on the fixed four-dimensional base, together with Kato's inequality, gives

$$\|s\|^2_{L^4(a)} \le C\big(2\mathfrak{q}_a(s,s) + \|s\|^2_a\big), \tag{44}$$

$$\int_{B_\rho(\Sigma)} |s|^2_a \,\mathrm{d}V \le C\rho^2\big(2\mathfrak{q}_a(s,s) + \|s\|^2_a\big). \tag{45}$$

The second estimate uses the $O(\rho^4)$ volume of the finitely many balls. Both are intrinsic estimates on the original smooth bundle and require no curvature bound.

Suppose $a_n \downarrow 0$, $\|s_n\|_{a_n} = 1$, and $\mathfrak{q}_{a_n}(s_n, s_n) \to 0$. Transport by F_{a_n} on fixed compact subsets of U. The convergence of the connection coefficients and the covariant derivative bound give local H^1 bounds. Rellich compactness, followed by a diagonal subsequence, gives strong local L^2 convergence to a section s_* satisfying $D_* s_* = 0$. It is smooth by elliptic regularity. Its holomorphic coefficients extend over the isolated points by Hartogs' theorem, and the extension is parallel. Estimate (45), first with fixed ρ and then with $\rho \downarrow 0$, shows that the limit has norm one. Applying the same argument to sums of finitely many sequences preserves their real inner products.

To obtain test sections, choose a cutoff χ_ρ equal to zero on $B_\rho(\Sigma)$, equal to one outside $B_{2\rho}(\Sigma)$, and satisfying $|\mathrm{d}\chi_\rho| \le C/\rho$. For $b \in \mathcal{P}$, extend

$$T_{a,\rho} b = \chi_\rho F_a b F_a^{-1}$$

by zero across the points. Its mass is exactly $\int \chi_\rho^2 |b|^2 \,\mathrm{d}V$. For fixed ρ, its energy tends as $a \downarrow 0$ to $\frac{1}{2}\int |\mathrm{d}\chi_\rho|^2 |b|^2 \,\mathrm{d}V \le C\rho^2 |b|^2$. These statements are uniform on the unit sphere of $\mathcal{P}$. Min–max, first with ρ fixed and then with $\rho \downarrow 0$, proves $\lambda_k(a) \to 0$. If the complementary bound in (41) failed, there would be $k+1$ orthonormal eigensections with vanishing energy. The preceding compactness argument would give $k+1$ orthonormal elements of $\mathcal{P}$, a contradiction.

For positivity at a fixed a, a zero mode would be a nonzero parallel Hermitian trace-free endomorphism. Its two eigenlines would split $V = A \oplus B$ holomorphically. Projecting ι to the factors gives sections of $A \otimes L^{-1}$ and $B \otimes L^{-1}$. If both are nonzero, their zero divisors have nonnegative Kähler degrees whose sum is

$$\deg A + \deg B - 2\deg L = \deg V - 2\deg L = 0.$$

Both sections are then nowhere zero, contrary to the nonempty zero scheme of ι. If only one projection is nonzero, its zero set would be a nonempty finite set; a line-bundle section on a smooth surface has either a divisor of zeros or no zeros. This is again impossible. Thus $\ker \mathcal{L}_a = 0$. This argument does not assert simplicity of V; a holomorphic nilpotent endomorphism in the repeated-factor case is not Hermitian and parallel.

If W is reduced, in each fixed coordinate ball

$$u_0 \asymp r^2, \qquad |\mathrm{d}u_0| \le Cr, \qquad |\beta|_0 \le Cr^{-3}.$$

For the last bound, the numerator in $\tau = H_0^{-1} q^* / (q H_0^{-1} q^*)$ is $O(r)$ and the denominator is comparable to r^2. Hence $\bar\partial \tau = O(r^{-2})$, and division by the inclusion, whose norm is comparable to r, gives the assertion. For $r \ge \sqrt{a}$, (43) is therefore bounded by Ca/r^3. Choose the cutoff at radius $\sqrt{a}$. Its derivative contributes $O(a)$ to the energy, and the connection term contributes at most

$$Ca^2 \int_{\sqrt{a}}^R r^{-6} r^3 \,\mathrm{d}r \le Ca.$$

The fixed exterior contributes $O(a^2)$, whereas the omitted mass is $O(a^2)$. Min–max proves $\lambda_k(a) \le Ca$.

For precision about the subspaces, let Π_a be the orthogonal projection onto the first k eigensections and let T_a be the just constructed cutoff map. With the full graded L^2 inner product on $\mathcal{P}$, the complementary gap gives

$$\|(I-\Pi_a)T_a b\|_a^2 \le Ca\|b\|_*^2, \qquad R_a^* R_a = I + O(a), \qquad R_a = \Pi_a T_a. \tag{46}$$

Thus R_a is an isomorphism onto the complete small spectral space. On fixed compact subsets the transported $T_a b$ equals b, so $R_a b \to b$ in local L^2. Expand $R_a b$ in a normalized low eigenbasis. Its coefficients are bounded, and the transported elliptic operators have uniformly smooth coefficients there. Interior estimates give convergence in every fixed C^j norm. For a general finite W, choose a sequence $\rho_j \downarrow 0$ and then thresholds $a_j \downarrow 0$ so that the fixed-ρ_j connection error is at most ρ_j^2 whenever $a \le a_j$. Using those cutoffs successively gives mass $I + o(1)$ and energy $o(1)$, and the same proof applies. No parameter regularity of that last cutoff choice is needed. □

5.2. **The normalized mean-curvature residual.** For the remainder of this section assume W is reduced and use Lemma 2.2. Choose a smooth positive weight ρ_a comparable to $\sqrt{a+|z|^2}$ on the fixed balls and comparable to one on their complement. For example one can take the square root of a plus a fixed smooth function equal to $|z|^2$ near each point and positive elsewhere. Define

$$\Phi_a = i\Lambda_\omega F_{H_a} - \kappa I, \qquad \ell(a) = 1 + |\log a|.$$

Proposition 5.2. *For each fixed nonnegative integer j,*

$$|D_a^j \Phi_a|_{H_a} \le C_j a \rho_a^{-2-j}. \tag{47}$$

Consequently

$$\|\Phi_a\|_p \le \begin{cases} C_p a, & 1 \le p < 2, \\ Ca\sqrt{\ell(a)}, & p = 2, \\ C_p a^{2/p}, & 2 < p < \infty, \end{cases} \qquad \|\Phi_a\|_\infty \le C. \tag{48}$$

The smooth parameter direction

$$k_a = H_a^{-1} a \partial_a H_a = \frac{2S - u_0 I}{u_0 + a} \tag{49}$$

satisfies

$$\|k_a\|_a^2 \longrightarrow 2v_0, \qquad \langle \Phi_a, k_a \rangle_a = \frac{2\mathcal{E}_a}{a} = 4\pi^2 \sigma a + o(a). \tag{50}$$

These assertions allow $L \simeq M$.

Proof. In the normalized frames near p, put $s = |z|^2$ and $c = \lambda_p \sqrt{h_M / h_L}$. Then

$$u_0 = cs, \quad u_a = \frac{acs}{a+cs}, \quad c = \lambda_p + O(r^2), \quad \mathrm{d}c = O(r), \quad g = g_{\mathrm{E}} + O(r^2).$$

The orthogonal quotient lift and its $\bar\partial$ derivative are

$$\tau = \lambda_p^{-1/2} \frac{(\bar z_2, \bar z_1)^{\mathsf{t}}}{s}, \qquad \beta = \frac{\bar z_1 \mathrm{d}\bar z_2 - \bar z_2 \mathrm{d}\bar z_1}{\lambda_p s^2}. \tag{51}$$

Since the line weight on $\mathrm{Hom}(M,L)$ is $(\lambda_p/c)^2$ and $|\mathrm{d}z_j|^2=2$,

$$|\beta|_0^2=\frac{2}{c^2s^3}+O(s^{-2}). \tag{52}$$

Write $\Delta=-\Delta_0$ for the nonpositive real scalar Laplacian. In the unitary splitting the diagonal residual is

$$(\Phi_a)_{11}=u_a^2|\beta|_0^2-\frac{1}{2}\Delta\log u_a,\qquad (\Phi_a)_{22}=-(\Phi_a)_{11}. \tag{53}$$

For example, the first identity follows directly by substituting (26) into $\Delta u_a=2e_a-2u_a(\Phi_a)_{11}$. The trace identity follows from the fixed HE determinant.

For $v=cs$ one has $\Delta v=8c+O(s)$ and $|\mathrm{d}v|^2=4c^2s+O(s^2)$. The derivatives of $f(v)=\log a+\log v-\log(v+a)$ are

$$f'(v)=\frac{a}{v(v+a)},\qquad f''(v)=-\frac{a(2v+a)}{v^2(v+a)^2}.$$

The chain rule therefore gives

$$\Delta\log u_a=\frac{4a^2}{s(a+cs)^2}+O\left(\frac{a}{a+s}\right), \tag{54}$$

$$u_a^2|\beta|_0^2=\frac{2a^2}{s(a+cs)^2}+O\left(\frac{a^2}{(a+s)^2}\right). \tag{55}$$

For the first error, multiply $O(s)$ by $|f'|\le Ca/[s(s+a)]$, and $O(s^2)$ by $|f''|\le Ca/[s^2(s+a)]$. The leading terms cancel in (53), leaving $Ca/(a+s)$.

The upper off-diagonal curvature block in the unnormalized splitting is

$$D_0'\beta+2\partial\log u_a\wedge\beta.$$

After unitary normalization its contraction has norm at most $Cu_a|\Lambda(D_0'\beta+2\partial\log u_a\wedge\beta)|_0$. Direct differentiation in the flat metric gives

$$\Lambda_{\mathrm{E}}\partial\beta=0,\qquad \Lambda_{\mathrm{E}}(\partial s\wedge\beta)=0.$$

The contraction error is $O(s)$, the fixed line connection is $O(r)$, and $\partial\beta=O(r^{-4})$. It follows that $|\Lambda D_0'\beta|_0\le Cs^{-1}$; multiplication by $u_a\le Cas/(a+s)$ gives the required bound. Finally

$$\partial\log u_a=\frac{a}{s(a+cs)}\partial s+\frac{a}{c(a+cs)}\partial c.$$

The flat contraction of the first term with β vanishes. The base-metric error and the second term each have norm at most $Ca/[s(a+s)]$ before multiplication by u_a. Their normalized contributions are at most $Ca^2/(a+s)^2$. Hermitian symmetry treats the other off-diagonal block. This proves (47) for $j=0$.

On $r\ge\sqrt{a}$, differentiating these formulas and their cancellations costs at most C_jr^{-j}. The connection in the normalized splitting is $O(r^{-1})$, with derivatives of order j bounded by C_jr^{-1-j}. Conversion to covariant derivatives preserves the estimate C_jar^{-2-j}. At the center use the smooth holomorphic frame instead. Under $z=\sqrt{a}\,y$, on a fixed y-ball,

$$H_a(\sqrt{a}\,y)=H^{\mathrm{flat}}_{\lambda_p}(y)+O_{C^j}(a),\qquad g(\sqrt{a}\,y)=g_{\mathrm{E}}+O_{C^j}(a).$$

The leading metric is the local HYM metric of Lemma 3.1, with a fixed dilation and sign change. Its metric and inverse are uniformly bounded on the fixed ball. The physical curvature contraction has a factor a^{-1}, while its leading term is zero and its error is $O_{C^j}(a)$. Thus the physical Φ_a and its y-covariant derivatives are bounded. Returning to physical

derivatives costs $a^{-j/2}$, which is precisely (47) on $r \le 2\sqrt{a}$. On the fixed exterior, (43) gives $O(a)$ in every fixed derivative. This proves the global estimate. Integrating

$$Ca^p \int_0^R \frac{r^3}{(a+r^2)^p}\,\mathrm{d}r$$

proves (48), including the logarithm at $p = 2$.

Differentiating the two eigenvalues w, w^{-1} in (27), with P independent of a, gives

$$k_a = \frac{u_0}{u_0 + a}(2P - I) = \frac{2S - u_0 I}{u_0 + a}.$$

The second expression is smooth across the points. Its eigenvalues are opposite, bounded in absolute value by one, and tend to ± 1 off Σ. Bounded convergence gives the mass limit in (50). Since Φ_a is trace-free,

$$\operatorname{tr}(\Phi_a k_a) = \frac{2}{a} u_a (\Phi_a)_{11}.$$

The integrated Bochner identity for the smooth global section ι says that the integral of $u_a(\Phi_a)_{11}$ is $\mathcal{E}_a$. There is no puncture boundary term. The balanced static energy (29) now proves the exact pairing and its asymptotic. □

The residual is small in each finite L^p space, but the estimate does not tend to zero in L^∞. A correction is therefore needed before H_a can provide a sufficiently accurate comparison for the flow. The complete dimension in (40) will also be retained in that correction.

6. A projected Hermitian–Einstein equation

Assume throughout this section that W is reduced. The factors L, M may still coincide. Let Π_a be the spectral projection onto the complete k-dimensional small space E_a from Proposition 5.1, and put $Q_a = I - \Pi_a$. We decrease a_0 so that the small eigenvalues are at most $c/4$ and the complementary spectrum is at least $c > 0$. For an H_a-Hermitian trace-free section v, define

$$\mathcal{M}_a(v) = e^{v/2}\big(i\Lambda F_{H_a e^v} - \kappa I\big)e^{-v/2}. \tag{56}$$

The conjugation is the fibre isometry from $H_a e^v$ to H_a; thus $\mathcal{M}_a(v)$ is H_a-Hermitian and trace-free. It is not a holomorphic change of the bundle.

6.1. **Scaled norms and the complementary inverse.** Fix $0 < \alpha < 1$ and a sufficiently small $\eta > 0$. On $B(x, 4\eta\rho_a(x))$, the weight ρ_a is comparable to $\rho_a(x)$. Rescale this ball by $\rho_a(x)^{-1}$. The rescaled base metrics are uniformly elliptic with bounded coefficients in every fixed smooth norm. There are also unitary frames with uniformly bounded scaled connection coefficients. On $r \le C\sqrt{a}$, use the smooth normalized holomorphic frame, the dilation $z = \sqrt{a}\,y$, and unitary normalization by $H_a^{-1/2}$. The matrix and its inverse have uniformly bounded derivatives on each fixed enlarged y-ball. On $r \ge \sqrt{a}$, use (42) and fixed unitary line frames. Their connection difference has derivatives $O(ar^{-3-j})$ of order j, by the explicit formulas in the proof of Proposition 5.2. Rescaling by r gives the claimed bounds. The fixed exterior is uniformly smooth. These descriptions overlap; none uses the singular splitting at a point of Σ.

Let X_a be the supremum of the rescaled $C^{2,\alpha}$ norms in these frames, including the global supremum norm. Let Y_a be the supremum of the rescaled $C^{0,\alpha}$ norms of $\rho_a(x)^2 f$ on the same

balls. Uniformly equivalent intrinsic norms are

$$\|v\|_{X_a} \asymp \|v\|_\infty + \sup_X\big(\rho_a|D_a v| + \rho_a^2|D_a^2 v|\big) + \sup_x \rho_a(x)^{2+\alpha}[D_a^2 v]_{\alpha;B(x,\eta\rho_a(x))}, \tag{57}$$

$$\|f\|_{Y_a} \asymp \sup_X \rho_a^2|f| + \sup_x \rho_a(x)^{2+\alpha}[f]_{\alpha;B(x,\eta\rho_a(x))}. \tag{58}$$

The Hölder differences can be taken after local parallel transport. For each fixed $a > 0$, these are ordinary Hölder spaces with equivalent norms. We do not differentiate a choice of charts when differentiating a section with respect to a. Proposition 5.2 and four-dimensional integration give

$$\|\Phi_a\|_{Y_a} + \|\Phi_a\|_{4/3} \le Ca, \qquad \|\rho_a^{-2}\|_1 + \|\rho_a^{-2}\|_{4/3} \le C. \tag{59}$$

In particular Y_a controls L^1 and $L^{4/3}$ uniformly.

Lemma 6.1. *The low projection and complementary inverse satisfy*

$$\|\Pi_a f\|_{X_a} \le C\|f\|_1, \qquad \|\Pi_a\|_{X_a\to X_a} + \|\Pi_a\|_{Y_a\to Y_a} \le C, \tag{60}$$

$$\|G_a f\|_{X_a} \le C\ell(a)\|f\|_{Y_a}, \qquad G_a = (\mathcal{L}_a|_{Q_a})^{-1}Q_a. \tag{61}$$

Proof. For the rough connection operator, scalar heat domination gives

$$|e^{-t\mathcal{L}_a} f|(x) \le e^{-t\Delta_0/2}|f|(x). \tag{62}$$

One may obtain this by applying the maximum principle to a regularized norm; there is no curvature potential in $\frac{1}{2}D_a^* D_a$. The semigroup and kernel forms are also given in [Gün12, Theorem 2.13] and [Gün11, Theorem 8.8(b), Remark 8.9], respectively. Complexification followed by restriction gives the real-bundle statement. On our fixed closed four-manifold the scalar kernel obeys

$$h_t(x,y) \le Ct^{-2}\exp\{-d(x,y)^2/(Ct)\}, \qquad 0 < t \le 1,$$

by the compact specialization of [LY86, Corollary 3.1]. The factor $1/2$ in (62) changes only constants.

If e is an L^2-normalized low eigensection, evaluation at heat time one gives $\|e\|_\infty \le C$. Uniform local estimates for its scaled elliptic equation give $\|e\|_{X_a} \le C$. Expansion in an orthonormal basis of the fixed-dimensional space E_a proves (60), using (59) and the boundedness of ρ_a.

We record the kernel bound needed for the weighted inverse. For $0 < t \le 1$, the kernel of $e^{-t\mathcal{L}_a}Q_a$ is the full kernel minus its k low terms, and hence has norm at most

$$Ct^{-2}\exp\{-d(x,y)^2/(Ct)\} + C.$$

For $t \ge 1$, place a fixed positive-time smoothing factor at both ends of the complementary semigroup. Heat domination bounds their L^2-to-fibre norms, while the middle factor is bounded by Ce^{-ct} by the gap. Integration in time therefore yields the fibre operator-norm bound

$$|G_a(x,y)| \le C(1 + d(x,y)^{-2}), \qquad x \ne y. \tag{63}$$

This is a bound in the current metrics, independent of any singular frame.

The four-dimensional convolution estimate

$$\sup_x \int_X \frac{1 + d(x,y)^{-2}}{\rho_a(y)^2}\,\mathrm{d}V_y \le C\ell(a) \tag{64}$$

now gives the supremum bound for $G_a f$. Here is a direct verification near a defect p. If $d(x,p) = O(\sqrt{a})$, the contribution within distance $O(\sqrt{a})$ of p is bounded by rescaling, and

the remaining radial integral is bounded by $\int_{\sqrt{a}}^{R} r/(a+r^2)\,\mathrm{d}r$. If $R_x = d(x,p)$ is larger, the radius-$R_x/2$ neighborhood of x has $\rho_a^2 \gtrsim a + R_x^2$ and bounded contribution. The radius-$R_x/2$ ball about p contributes at most

$$CR_x^{-2}\int_0^{R_x/2}\frac{r^3}{a+r^2}\,\mathrm{d}r \le C.$$

The intervening comparable annulus away from x also contributes a constant. On $r > 2R_x$, $d(x,y) \gtrsim r$ and the same logarithmic integral applies. The fixed exterior is uniformly integrable. Summing over the finitely many points proves (64). Finally, the uniform scaled Schauder estimate for $\mathcal{L}_a G_a f = Q_a f$, together with (60), upgrades the supremum estimate to (61). □

6.2. **Solving the projected equation.**

Theorem 6.2. *There are fixed $K, a_0 > 0$ such that for $0 < a \le a_0$ there is a unique $v_a \in Q_a X_a$ in the ball $\|v_a\|_{X_a} \le Ka\ell(a)$ satisfying*

$$Q_a \mathcal{M}_a(v_a) = 0. \tag{65}$$

The metric $\widehat{H}_a = H_a e^{v_a}$ is smooth and has the prescribed determinant. Moreover

$$\|v_a\|_{X_a} \le Ca\ell(a), \qquad \|D_a v_a\|_2 \le Ca, \qquad \|\Phi_{\widehat{H}_a}\|_\infty \le Ca, \tag{66}$$

$$\widehat{Z}_a \sim a v_0, \qquad \widehat{\mathcal{E}}_a = 2\pi^2\sigma a^2 + o(a^2). \tag{67}$$

Uniqueness is asserted only in the indicated ball and slice.

Proof. The exact finite metric-change formula is

$$F_{H_a e^v} = F_{H_a} + \bar\partial_{\mathrm{End}\,V}\big(e^{-v}D_a'(e^v)\big).$$

Differentiating (56) at zero gives

$$i\Lambda\bar\partial D_a' v + \tfrac{1}{2}[v,\Phi_a] = \tfrac{1}{2}\big((D_a')^*D_a' + (D_a'')^*D_a''\big)v = \mathcal{L}_a v.$$

This is the Kähler identity on endomorphism-valued zero-forms: the centering commutator cancels half the difference between the two Dolbeault Laplacians. See [Kob87, (3.2.39)–(3.2.41), (6.3.16)–(6.3.17)] for the commutators and curvature variation. In particular the real linearization is valid even though Φ_a is not central.

Write

$$\mathcal{M}_a(v) = \Phi_a + \mathcal{L}_a v + N_a(v). \tag{68}$$

On a fixed sufficiently small X_a ball, the finite change formula and the uniform scaled coefficients imply

$$\|N_a(v) - N_a(w)\|_{Y_a} \le C(\|v\|_{X_a} + \|w\|_{X_a})\|v-w\|_{X_a}. \tag{69}$$

Indeed, in a unit rescaled chart the centered curvature is a smooth matrix expression of order two with uniformly bounded coefficients. After subtracting its value and first derivative, every term contains at least two factors from v and its derivatives, and at most two total spatial derivatives. The $C^{0,\alpha}$ product estimate proves (69); multiplication by ρ_a^2 converts physical curvature to that expression. Terms involving $vD_a^2 v$ are included in this estimate.

On $Q_a X_a$, equation (65) becomes

$$v = -G_a\big(\Phi_a + N_a(v)\big).$$

By (59), (61) and (69), its right side has norm at most $C\ell(a)(a + \|v\|_{X_a}^2)$. Its Lipschitz constant on the radius-$Ka\ell(a)$ ball is at most $CKa\ell(a)^2$. Choose K larger than twice the linear bound and then decrease a_0. The contraction theorem gives the claimed solution and local uniqueness. For fixed a, the right side of the unprojected equation is a linear combination of smooth low eigensections. Its principal coefficients are a small perturbation of those of $\mathcal{L}_a$. Quasilinear elliptic regularity makes v_a smooth. Hermitian symmetry gives positivity of $H_a e^{v_a}$, and $\operatorname{tr} v_a = 0$ fixes its determinant.

The remaining centered curvature is

$$\mathcal{M}_a(v_a) = \Pi_a \Phi_a + \Pi_a N_a(v_a).$$

The L^1-to-X_a projection estimate gives its supremum norm at most $C(a + a^2\ell(a)^2) \le Ca$. Centering preserves the fibre norm, so this proves the curvature estimate in (66).

For the stronger gradient bound, pair the equation with v_a. The $L^{4/3}$ bound in (59), the same bound for the quadratic remainder, the uniform Sobolev inequality (44) and the complementary gap give

$$\mathfrak{q}_a(v_a, v_a) \le \|\Phi_a + N_a(v_a)\|_{4/3} \|v_a\|_4 \le Ca\, \mathfrak{q}_a(v_a, v_a)^{1/2}.$$

Here $\|N_a(v_a)\|_{4/3} \le Ca^2\ell(a)^2$, and on Q_a the L^2 term in the Sobolev estimate is controlled by $\mathfrak{q}_a$. Thus $\|D_a v_a\|_2 \le Ca$.

The relative metric has supremum logarithm $O(a\ell(a))$; hence $\widehat{Z}_a/Z_a = 1 + O(a\ell(a))$. For the energy, the finite connection-change formula gives pointwise

$$|(D'_{\widehat{H}_a} - D'_{H_a})\iota| \le C\sqrt{u_a}\, |D_a v_a|.$$

Since $u_a \le a$, its L^2 norm is $O(a^{3/2})$. Also $\|D'_{H_a \otimes h_L^{-1}} \iota\|_2 = O(a)$. Comparing the squared norms and the two fibre metrics gives

$$|\widehat{\mathcal{E}}_a - \mathcal{E}_a| \le Ca^{5/2} + Ca^3 + Ca^3\ell(a) = o(a^2).$$

The balanced identities (29) now prove (67). □

The correction uses the whole small spectral space, including all three directions when the line factors repeat. The remaining curvature is not yet identified with the derivative of the metric family. That identification requires estimates with respect to the parameter, which we prove next.

7. Parameter estimates and a comparison trajectory

The correction in Section 6 was obtained at each fixed a. We now control its parameter derivative. The derivative estimates still allow repeated line factors. The scalar trajectory constructed at the end of this section requires $L \not\simeq M$.

7.1. Differentiation in a fixed unitary bundle.

Write $H_a = H_0 T_a$, with T_a as in (23). The isometry $U_a = T_a^{1/2} : (V, H_a) \to (V, H_0)$ satisfies $a\partial_a U_a = \frac{1}{2} U_a k_a$, since T_a and its parameter derivative commute. For a moving Hermitian section define its derivative in this fixed unitary identification by

$$\dot{s} = a\partial_a s + \tfrac{1}{2}[k_a, s]. \tag{70}$$

A dot on an operator has the same meaning. The base volume and the Hilbert inner product in the identified bundle are fixed.

Lemma 7.1. *The explicit family satisfies*

$$\|k_a\|_{X_a} \le C, \qquad \|\dot\Phi_a\|_{Y_a} \le Ca, \qquad \|\dot{\mathcal{L}}_a\|_{X_a\to Y_a} \le C. \tag{71}$$

Moreover

$$\dot\Phi_a = \mathcal{L}_a k_a, \qquad \|Q_a k_a\|_{X_a} \le Ca\ell(a). \tag{72}$$

Proof. On the annuli the identification by U_a fixes the unitary graded frame: $U_a F_a = F_0$, because its two multipliers are

$$w^{1/2}u_a^{-1/2} = u_0^{-1/2}, \qquad w^{-1/2}u_a^{1/2} = u_0^{1/2}.$$

Thus ordinary parameter differentiation of the normalized graded matrices is exactly (70). Applying $a\partial_a$ to the explicit rational factors with denominator $a+u_0$ preserves their symbol estimates. In particular, the cancelled residual formulas in Proposition 5.2, including their spatial derivatives, remain $O(ar^{-2-j})$. The expression (49) similarly has uniformly bounded scaled derivatives. This argument differentiates the formulas and their cancellations, not an unspecified remainder estimate.

On the core use $z = \sqrt{a}\,y$. The base and line data have vanishing first jets. Their integral second-order Taylor remainders, and every fixed number of y derivatives, are $O(a)$, also after a logarithmic derivative at fixed y. Differentiation at fixed z is $a\partial_a - \frac{1}{2}y\cdot\partial_y$ in these expressions. Both terms preserve the bounds on a fixed enlarged ball. The flat leading contracted curvature is identically zero. Its physical residual and its logarithmic parameter derivative are therefore bounded in every fixed y norm. The unitary normalization and its derivative are uniformly bounded there. This proves the first two estimates in (71) on the core as well. The fixed exterior follows directly from smooth formulas.

The unitary variation of the Chern connection is the one-form

$$B_a = \tfrac{1}{2}(D_a' k_a - D_a'' k_a),$$

acting by commutator on endomorphisms. It follows by differentiating the Chern change formula and U_a. The bounds just proved give $|D_a^j B_a| \le C_j \rho_a^{-1-j}$. Differentiation of $\frac{1}{2}D_a^* D_a$ has first-order terms with coefficient B_a and zeroth-order terms with one derivative of B_a. The scaled Hölder coefficient bounds prove the third estimate in (71). There is no derivative of the inner product in this unitary identification. Finally the centered linearization in the preceding section, applied to the metric velocity k_a, gives $\dot\Phi_a = \mathcal{L}_a k_a$. Apply G_a and (61) to obtain the second assertion in (72). □

Proposition 7.2. *The family v_a is smooth for $a>0$, and*

$$\|\dot v_a\|_{X_a} \le Ca\ell(a)^3. \tag{73}$$

If the velocity of $\widehat H_a$ is centered to H_a by

$$\widehat k_a^{\,c} = e^{v_a/2}\big(\widehat H_a^{-1} a\partial_a \widehat H_a\big)e^{-v_a/2},$$

then

$$\|\widehat k_a^{\,c} - k_a\|_\infty \le Ca\ell(a)^3. \tag{74}$$

Proof. Use the fixed resolvent circle $|z| = c/2$. Its distance from both the low spectrum and the complementary spectrum is bounded below. On that circle,

$$\|(z-\mathcal{L}_a)^{-1}\|_{Y_a\to X_a} \le C\ell(a).$$

For the low summands this follows from (60). On the complement, multiply the heat representation by e^{zt}. Its short-time bound is unchanged, and its large-time tail still decays

exponentially because $\operatorname{Re} z \le c/2$. The kernel and scaled Schauder arguments in Lemma 6.1 apply. The resolvent derivative and (71) consequently give

$$\|\dot\Pi_a\|_{Y_a\to X_a} \le C\ell(a)^2. \tag{75}$$

Both resolvents map Y_a to X_a; the middle operator derivative maps X_a to Y_a. The bounded embedding $X_a \to Y_a$ permits use of the same estimate on X_a. Differentiating $\Pi_a^2 = \Pi_a$ shows that $\dot\Pi_a$ exchanges the low and complementary spaces; in particular $\Pi_a\dot\Pi_a\Pi_a = 0$. No separation of eigenvalues within the low cluster is required.

With v held fixed in the unitary identification, the parameter derivative of the quadratic remainder obeys

$$\|\dot N_a^{\mathrm{par}}(v)\|_{Y_a} \le C\|v\|_{X_a}^2. \tag{76}$$

To see this, differentiate the finite curvature formula after subtracting its value and first Fréchet derivative. The scaled coefficients and their required parameter derivatives are uniformly bounded by the explicit annular and core calculations above. The second Fréchet derivative, and its parameter derivative, are therefore uniformly bounded on a fixed small ball. Matrix Taylor's formula proves (76). The estimate includes second-jet terms. The logarithmic derivative of ρ_a is bounded, so the statement is in the same scaled norms at the parameter under consideration.

The complementary derivative of the equation is invertible:

$$\Big\|\big((\mathcal{L}_a + Q_aDN_a(v_a))|_{Q_aX_a}\big)^{-1}\Big\|_{Q_aY_a\to Q_aX_a} \le C\ell(a). \tag{77}$$

Indeed the perturbation times G_a is $O(a\ell(a)^2)$, so a Neumann series applies. On compact positive parameter intervals, the projection and coefficients are smooth. A local smooth identification of Q_a makes the ordinary implicit function theorem applicable. Uniqueness in Theorem 6.2 identifies the local solutions and proves smoothness for $a > 0$.

Put $\mathcal{F}_a(v) = \Phi_a + N_a(v)$. Differentiating $\Pi_a v_a = 0$ gives

$$\Pi_a\dot v_a = -\dot\Pi_a v_a, \qquad \|\Pi_a\dot v_a\|_{X_a} \le Ca\ell(a)^3.$$

Differentiating $\mathcal{L}_a v_a + Q_a\mathcal{F}_a(v_a) = 0$ and projecting onto Q_a gives

$$\begin{aligned}(\mathcal{L}_a + Q_aDN_a(v_a))Q_a\dot v_a = {}& -Q_a\dot{\mathcal{L}}_a v_a + Q_a\dot\Pi_a\mathcal{F}_a(v_a) - Q_a\dot\Phi_a\\ & - Q_a\dot N_a^{\mathrm{par}}(v_a) - Q_aDN_a(v_a)\Pi_a\dot v_a.\end{aligned}$$

The five terms have Y_a norms

$$O(a\ell(a)),\quad O(a\ell(a)^2),\quad O(a),\quad O(a^2\ell(a)^2),\quad O(a^2\ell(a)^4),$$

respectively. Their sum is $O(a\ell(a)^2)$. Applying (77) proves (73).

For the tangent, differentiation of the exponential yields the exact identity

$$\widehat{k}_a^{\,c} = \cosh(\mathrm{ad}_{v_a}/2)k_a + \frac{\sinh(\mathrm{ad}_{v_a}/2)}{\mathrm{ad}_{v_a}/2}\,\dot v_a. \tag{78}$$

Substitute $a\partial_a v_a = \dot v_a - \frac{1}{2}[k_a, v_a]$ in the integral derivative formula for e^{v_a}; the two k_a terms combine to the displayed hyperbolic cosine. The quotient is defined by its entire power series at zero. Its operator norm is bounded on the small supremum ball, and the cosine differs from the identity by $O(|v_a|^2)$. Estimates (66) and (73) prove (74). □

7.2. **The single small eigenvalue.** From now on assume $L \not\cong M$. Set

$$e_a = \Pi_a k_a, \qquad b_0 = \frac{2\pi^2\sigma}{v_0}. \tag{79}$$

The section e_a is nonzero for small a: its mass tends to $2v_0$ by (50) and (72).

Proposition 7.3. *There is a smooth positive scalar $b(a)$, for a sufficiently small, such that*

$$\mathcal{M}_a(v_a) = b(a)e_a, \qquad b(a) = b_0 a + o(a). \tag{80}$$

For the real connection operators of H_a and $\widehat{H}_a$,

$$\lambda_1(\mathcal{L}_a) = b_0 a + o(a), \qquad \lambda_1(\widehat{\mathcal{L}}_a) = b_0 a + o(a), \qquad \lambda_2(\widehat{\mathcal{L}}_a) \geq c' > 0. \tag{81}$$

Proof. The remaining centered curvature belongs to the one-dimensional space spanned by e_a, defining $b(a)$. Pairing with k_a and using self-adjointness gives

$$\begin{aligned} b(a)\|e_a\|_2^2 &= \langle \Phi_a, k_a\rangle + \langle v_a, \mathcal{L}_a k_a\rangle + \langle N_a(v_a), k_a\rangle \\ &= 4\pi^2\sigma a + o(a) + O(a^2\ell(a)) + O(a^2\ell(a)^2). \end{aligned}$$

Here $\|\mathcal{L}_a k_a\|_1 = \|\dot{\Phi}_a\|_1 = O(a)$. Since $\|e_a\|_2^2 \to 2v_0$, this proves (80), including positivity.

We compute the eigenvalue separately. Put $J = 2P - I$ and $\zeta = u_0/(u_0 + a)$, so $k_a = \zeta J$ and $u_a = a\zeta$. In the unitary splitting, the diagonal derivative of ζ and off-diagonal derivative of J are orthogonal. Hence

$$|D_a'' k_a|^2 = \frac{2a^2|\partial u_0|^2 + 4a^2u_0^4|\beta|_0^2}{(u_0+a)^4} = \frac{4a^2u_0e_0 - 2a^2|\partial u_0|^2}{(u_0+a)^4}. \tag{82}$$

The second equality uses (26) for H_0. At a normalized reduced point,

$$u_0 = \lambda_p r^2 + O(r^4), \qquad e_0(p) = 4\lambda_p, \qquad |\partial u_0|^2 = 2\lambda_p^2 r^2 + O(r^4).$$

The density in (82) is nonnegative and bounded by $Ca^2r^2/(a+r^2)^4$. After $z = \sqrt{a}\,y$ and division of its integral by a, the dominating function is $C|y|^2/(1+|y|^2)^4$, integrable in four real dimensions. The leading integral at p is

$$12\pi^2a^2\lambda_p^2 \int_0^\infty \frac{s^2}{(a+\lambda_p s)^4}\,\mathrm{d}s = \frac{4\pi^2 a}{\lambda_p},$$

since $\int_0^\infty t^2(1+t)^{-4}\,\mathrm{d}t = 1/3$. The fixed exterior is $O(a^2)$. Thus $\mathfrak{q}_a(k_a,k_a) = 4\pi^2\sigma a + o(a)$. The complementary energy is

$$\mathfrak{q}_a(Q_a k_a, Q_a k_a) = \langle Q_a k_a, \mathcal{L}_a k_a\rangle = O(a^2\ell(a)) = o(a).$$

As e_a spans the first eigenspace and $\|e_a\|_2^2 \to 2v_0$, this proves the reference eigenvalue equivalent.

Under the isometry $e^{v_a/2}$, the corrected unitary connection differs from D_a by a one-form A_a with $|A_a| \leq C|D_a v_a|$. The weighted derivative bound gives

$$\|A_a\|_4 \leq Ca\ell(a)^{5/4}, \qquad \int_X \rho_a^{-4}\,\mathrm{d}V \leq C\ell(a).$$

For every real Hermitian trace-free section s, the uniform Sobolev inequality yields

$$\|A_a s\|_2 \leq Ca\ell(a)^{5/4}\big(\mathfrak{q}_a(s,s) + \|s\|_2^2\big)^{1/2}.$$

Expansion of the corrected connection energy and the product inequality with parameter $\sqrt{a}$ imply

$$|\widehat{\mathfrak{q}}_a(s,s) - \mathfrak{q}_a(s,s)| \leq C\sqrt{a}\,\mathfrak{q}_a(s,s) + Ca^{3/2}\ell(a)^{5/2}\|s\|_2^2. \tag{83}$$

A term of order $a^{3/2}\ell(a)^{5/2}\mathfrak{q}_a$ is absorbed into the first term for small a. The mass forms are identical under the isometry. Min–max, the reference equivalent and its complementary gap prove the remaining assertions in (81). □

7.3. **A trajectory with integrable flow error.** Choose a sufficiently small fixed $a_{\mathrm{in}} > 0$ and solve

$$a'(T) = -2a(T)b(a(T)), \qquad a(0) = a_{\mathrm{in}}. \tag{84}$$

Define $\overline{H}(T) = \widehat{H}_{a(T)}$. This is a curve of smooth metrics on the original compact bundle.

Proposition 7.4. *The curve exists for all $T \geq 0$, and*

$$a(T) \sim \frac{1}{2b_0 T}, \qquad \overline{Z}(T) \sim \frac{v_0^2}{4\pi^2 \sigma T}. \tag{85}$$

Its metric-flow error

$$\mathcal{R}(T) = \overline{H}^{-1}\partial_T \overline{H} + 2\Phi_{\overline{H}}$$

obeys

$$\|\mathcal{R}(T)\|_{\infty,\overline{H}} \leq Ca(T)^2\ell(a(T))^3, \qquad \int_T^\infty \|\mathcal{R}(t)\|_\infty \,\mathrm{d}t \leq Ca(T)\ell(a(T))^3. \tag{86}$$

Proof. For small a, $b(a)/a$ lies between two positive constants. The solution stays positive, decreases to zero and exists for all nonnegative time. Its reciprocal satisfies

$$(1/a)' = 2b(a)/a \longrightarrow 2b_0.$$

Integration proves the first equivalent in (85); the corrected mass in (67) proves the second. Centering the error gives the exact expression

$$e^{v_a/2}\mathcal{R}e^{-v_a/2} = 2b(a)\big(e_a - \widehat{k}_a^c\big).$$

The term in parentheses has supremum norm $O(a\ell(a)^3)$ by (72) and (74). Since $b(a) = O(a)$, this proves the first estimate in (86). For the second, substitute $\mathrm{d}T = -\mathrm{d}a/(2ab(a))$ and use $b(a) \geq ca$. The tail is bounded by

$$C\int_0^{a(T)} (1 + |\log x|)^3 \,\mathrm{d}x \leq Ca(T)\ell(a(T))^3.$$

□

The coefficient and time scale in this proposition belong so far to the constructed curve. The next section proves convergence of every actual flow with the fixed determinant to that curve. In particular, integrability of the error alone will first give bounded metric distance; the small positive eigenvalue supplies the additional decay.

8. Comparison with the actual flow

Assume that W is reduced and $L \not\simeq M$. Let $H(T)$ solve (10) from any smooth initial metric with determinant h_{det}, and let $\overline{H}(T) = \widehat{H}_{a(T)}$ be the curve in Proposition 7.4. No smallness condition is imposed on the initial distance between them.

Theorem 8.1. *There are $C > 0$ and $0 < \gamma < 1$, depending on the fixed data and the initial metric, such that*

$$\sup_X \left|\log\bigl(\overline{H}(T)^{-1}H(T)\bigr)\right|_{\overline{H}(T)} \le C(1+T)^{-\gamma}. \tag{87}$$

Consequently

$$Z(T) \sim \frac{v_0^2}{4\pi^2\sigma T}. \tag{88}$$

With the intrinsic amplitude $\alpha(T) = Z(T)/v_0$,

$$\sup_X \left|\frac{u(T)}{Z(T)} - \frac{1}{v_0}\frac{u_0}{u_0+\alpha(T)}\right| \le C(1+T)^{-\gamma}. \tag{89}$$

The same estimate holds with $a(T)$ in place of $\alpha(T)$. There are fixed $r_0, T_0 > 0$ such that for $0 < r \le r_0$ and $T \ge T_0$,

$$\sup_{X\setminus B_r(\Sigma)} \left|\frac{u(T)}{Z(T)} - \frac{1}{v_0}\right| \le C\left(T^{-\gamma} + \frac{1}{Tr^2}\right). \tag{90}$$

Proof. At each point the metrics with fixed determinant form the positive-matrix symmetric space with tangent norm $\operatorname{tr}((H^{-1}\delta H)^2)$. Put

$$s = \log(\overline{H}^{-1}H), \qquad H_\theta = \overline{H}e^{\theta s}, \qquad d = |s|_{\overline{H}}, \qquad e = \tfrac{1}{2}d^2.$$

The endomorphism s is trace-free and Hermitian for every H_θ. The metric-distance calculation for two solutions of HYM flow, in the convention (10), is

$$(\partial_T + \Delta_0)e = -2\int_0^1 |D'_{H_\theta}s|^2_{H_\theta}\,\mathrm{d}\theta.$$

This is [Kob87, Lemma 6.4.5], with book time $t = 2T$ and scalar operator $\Delta_0/2$. For our second endpoint the additional velocity is $\mathcal{R} = \overline{H}^{-1}\overline{H}_T + 2\Phi_{\overline{H}}$. The first variation of squared symmetric-space distance has endpoint gradient of norm d. Thus, with $\varepsilon(T) = \|\mathcal{R}(T)\|_\infty$,

$$(\partial_T + \Delta_0)e \le -2\int_0^1 |D'_{H_\theta}s|^2_{H_\theta}\,\mathrm{d}\theta + \varepsilon(T)d. \tag{91}$$

This uses only the endpoint velocity, with no derivative of $\mathcal{R}$.

For every θ, metric compatibility and Kato's inequality give

$$|\nabla d|^2 \le |D_{H_\theta}s|^2 = 2|D'_{H_\theta}s|^2.$$

Average over θ and apply the chain rule to (91). The derivative term has the sign needed to give

$$(\partial_T + \Delta_0)d \le \varepsilon(T). \tag{92}$$

At $d = 0$ use $(d^2+\eta^2)^{1/2}$ and then $\eta \downarrow 0$. The maximum principle and (86) imply

$$\sup_X d(T) \le B := \sup_X d(0) + \int_0^\infty \varepsilon(t)\,\mathrm{d}t < \infty. \tag{93}$$

Every smooth positive initial metric on the compact bundle gives a finite B.

We use the first eigenvalue in (81) to improve boundedness to decay. Choose a_{in} small enough that $\lambda_1(\widehat{\mathcal{L}}_a) \ge c_1 a$ throughout the comparison trajectory. The tensor $D''s$ is the same for all H_θ, because the holomorphic structure is fixed. Its squared fibre norms differ by

factors between e^{-2B} and e^{2B}. For example, after diagonalizing s at a point, a matrix entry has factor $e^{\theta(s_i-s_j)}$. As the D' and D'' norms of a Hermitian section agree, it follows that

$$\int_X |D'_{H_\theta} s|^2 \,\mathrm{d}V \geq e^{-2B}\widehat{\mathfrak{q}}_{a(T)}(s,s) \geq e^{-2B}c_1 a(T) \int_X d^2 \,\mathrm{d}V.$$

Put $D(T) = \|d(T)\|_2$. Integration of (91) and Cauchy–Schwarz give, with the usual regularization at $D = 0$,

$$D'(T) \leq -2e^{-2B}c_1 a(T) D(T) + \sqrt{v_0}\,\varepsilon(T). \tag{94}$$

Here $a(T) \asymp (1+T)^{-1}$, while

$$\varepsilon(T) \leq C(1+T)^{-2}(1+\log(2+T))^3.$$

Choose $0 < \gamma < 1$ smaller than an eventual coefficient of $(1+T)^{-1}$ in the damping term. Multiplying (94) by $(1+T)^\gamma$ and integrating gives $D(T) \leq C(1+T)^{-\gamma}$, since $\int_1^\infty t^{\gamma-2}(1+\log t)^3 \,\mathrm{d}t < \infty$. The fixed scalar heat comparison for (92) on $[T-1, T]$ now gives

$$\|d(T)\|_\infty \leq CD(T-1) + \int_{T-1}^{T} \varepsilon(t) \,\mathrm{d}t \leq C(1+T)^{-\gamma}.$$

This proves (87).

The correction from $H_{a(T)}$ to $\overline{H}(T)$ has supremum logarithm $O(a\ell(a))$. Decreasing γ if needed, the preceding estimate yields the relative quadratic-form bounds

$$e^{-CT^{-\gamma}} H_{a(T)} \leq H(T) \leq e^{CT^{-\gamma}} H_{a(T)}. \tag{95}$$

Evaluation on the fixed inclusion and integration give

$$u(T,x) = u_{a(T)}(x)(1+O(T^{-\gamma})), \qquad Z(T) = Z_{a(T)}(1+O(T^{-\gamma})),$$

with uniform errors off Σ; both inclusion norms vanish on Σ. For reduced W, (29) and (24) say

$$Z_a = av_0 + O(a^2), \qquad u_a = \frac{au_0}{u_0 + a}.$$

Together with (85), these prove (88) and the profile with parameter $a(T)$. They also give $\alpha(T)/a(T) = 1 + O(T^{-\gamma})$. For $u_0 \geq 0$, changing a to α in $u_0/(u_0+a)$ changes the result by at most $C|\alpha/a - 1|$, when this ratio is bounded above and below. This proves (89).

Near the reduced zeros $u_0 \asymp r^2$, and u_0 is bounded below on the fixed exterior. Hence $u_0 \geq cr^2$ on $X \setminus B_r(\Sigma)$. The error in replacing $u_0/(u_0+a)$ by one is at most $a/u_0 \leq C/(Tr^2)$, proving (90). □

In particular, a sufficient time for an exterior error at most ϵ, with constants depending only on the specified data and initial metric, is

$$T \geq C \max\{1, \epsilon^{-1/\gamma}, (\epsilon r^2)^{-1}\}.$$

This conclusion has joint time and radius quantifiers. It follows from the global comparison, rather than from qualitative convergence on a fixed compact set.

8.1. **The intrinsic scale of the inclusion.** For each reduced point put

$$r_p(T) = \sqrt{\frac{\alpha(T)}{\lambda_p}}. \tag{96}$$

In the fixed normalized holomorphic coordinates and frames, define

$$H_{\mathrm{loc}}(y) = (1+|y|^2)I - \frac{|y|^2+2}{1+|y|^2}(y_1,-y_2)^{\mathsf{t}}(\bar{y}_1,-\bar{y}_2). \tag{97}$$

This is the metric of Lemma 3.1 at parameter one, with the fixed unitary sign change.

Corollary 8.2. *As $T\to\infty$, uniformly on each compact y-set,*

$$H(T,r_p(T)y)\longrightarrow H_{\mathrm{loc}}(y), \qquad \frac{u(T,r_p(T)y)}{\alpha(T)}\longrightarrow \frac{|y|^2}{1+|y|^2}. \tag{98}$$

Along each radial ray at p, the first radius where $u(T,\cdot)=\alpha(T)/2$ is $r_p(T)(1+o(1))$, uniformly in direction.

Proof. The normalized coordinates give $u_0(z)=c(z)|z|^2$ with $c(z)=\lambda_p+O(|z|^2)$. Thus $u_0(r_py)/\alpha\to|y|^2$ smoothly on compact sets. Since $\alpha/a\to1$, the exact compact metric formula converges there to (97); its determinant weight tends to one. The limit and its inverse are bounded on each compact set. The relative estimate (95) therefore proves the first assertion in (98). Evaluation on $\iota(r_py)/\sqrt{\alpha}=(y_1,-y_2)^{\mathsf{t}}$ proves the second. For $0<\delta<1$, the scalar limit is strictly below $1/2$ on $|y|\le1-\delta$, and strictly above $1/2$ on $|y|=1+\delta$. For all sufficiently large T the same statements hold uniformly for the actual scalar norm. Every ray has a crossing between these two radii, and none before the smaller one. Letting $\delta\downarrow0$ proves the claimed first-crossing equivalent. □

The metric convergence just proved is in C^0. It does not yet imply convergence of curvature, and the radius defined by the inclusion need not agree by definition with any curvature radius. We establish the higher regularity and compare the two radii in the next section.

9. Derivative energy and curvature concentration

We retain the reduced, distinct-factor hypotheses of Theorem 8.1. All assertions below concern the actual solution $H(T)$.

Theorem 9.1. *For sufficiently large T,*

$$\|\Phi_{H(T)}\|_\infty\le\frac{C}{1+T}, \qquad \frac{c}{(1+T)^2}\le\mathcal{E}(T)\le\frac{C}{(1+T)^2}. \tag{99}$$

At each $p\in\Sigma$, the convergence

$$H(T,r_p(T)y)\longrightarrow H_{\mathrm{loc}}(y)$$

holds in C^∞ on every compact y-set, in the fixed holomorphic frames of Lemma 2.2. In particular, for every fixed $R>0$,

$$\int_{B_{Rr_p(T)}(p)}|F_{H(T)}|^2\,\mathrm{d}V\longrightarrow8\pi^2\mathfrak{F}(R), \qquad \mathfrak{F}(R)=\frac{R^4(R^2+3)}{(1+R^2)^3}. \tag{100}$$

The first radius $r_{F,p}(T)$ where the curvature energy in the coordinate ball about p reaches $4\pi^2$ satisfies

$$r_{F,p}(T)=r_p(T)(1+o(1)). \tag{101}$$

9.1. **The Donaldson functional and mean-curvature decay.** Use the Donaldson functional with an arbitrary additive constant and the first variation

$$\mathrm{d}\mathcal{D}_H(Hs) = \int_X \mathrm{tr}(\Phi_H s)\,\mathrm{d}V, \qquad \mathrm{d}V = \omega^2/2. \tag{102}$$

The first and second variations are given in [Kob87, (6.3.27), (6.3.29)–(6.3.34)]. On a metric geodesic $H_\theta = He^{\theta s}$, its acceleration is zero, so the general second-variation formula and integration by parts give

$$\frac{\mathrm{d}^2}{\mathrm{d}\theta^2}\mathcal{D}(H_\theta) = \|D'_{H_\theta} s\|_2^2 \geq 0. \tag{103}$$

No HE endpoint is needed for this calculation. The actual flow in our time convention satisfies

$$\frac{\mathrm{d}}{\mathrm{d}T}\mathcal{D}(H(T)) = -2\|\Phi_{H(T)}\|_2^2. \tag{104}$$

All three identities use the same volume normalization.

Lemma 9.2. *The functional has a finite infimum $\mathcal{D}_{\inf}$ on the smooth metrics of determinant $h_{\det}$, and*

$$0 \leq \mathcal{D}(H(T)) - \mathcal{D}_{\inf} \leq C/T \tag{105}$$

for large T.

Proof. The corrected family has $\|\Phi_{\widehat{H}_a}\|_\infty \leq Ca$ and bounded logarithmic-parameter velocity, by (66) and (74). Thus (102) gives $|\partial_a \mathcal{D}(\widehat{H}_a)| \leq C$. There is a finite limit $\mathcal{D}_{\inf}$, with

$$|\mathcal{D}(\widehat{H}_a) - \mathcal{D}_{\inf}| \leq Ca.$$

For a fixed smooth determinant-fixed metric H, the supremum distance to $\widehat{H}_a$ is at most $C_H + C|\log a|$. Indeed the reference eigenvalues w, w^{-1} lie between fixed positive multiples of a and a^{-1}, and the correction is $O(a\ell(a))$. Convexity at $\widehat{H}_a$ gives

$$\mathcal{D}(H) \geq \mathcal{D}(\widehat{H}_a) + \int_X \mathrm{tr}\big(\Phi_{\widehat{H}_a} \log(\widehat{H}_a^{-1} H)\big)\,\mathrm{d}V.$$

The final term is $O_H(a\ell(a))$. Letting $a \downarrow 0$ with H fixed proves that $\mathcal{D}(H) \geq \mathcal{D}_{\inf}$. The comparison metrics attain this value in the limit, so it is the infimum.

For any fixed smooth metric K, let $d_2(H,K)^2 = \int_X |\log(H^{-1}K)|_H^2\,\mathrm{d}V$. The endpoint variation of squared distance and convexity give

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t} d_2(H(t),K)^2 \leq 2\{\mathcal{D}(K) - \mathcal{D}(H(t))\}. \tag{106}$$

To check the sign and factor, the gradient of half squared distance at H is minus the logarithm toward K; the velocity is $-2\Phi_H$; and convexity bounds the pairing of Φ_H with that logarithm by $\mathcal{D}(K) - \mathcal{D}(H)$.

For each final T, fix $K = \widehat{H}_{a(T)}$ while integrating (106) over $[T/2, T]$. The initial distance in that interval is uniformly bounded: Theorem 8.1 compares $H(T/2)$ with $\widehat{H}_{a(T/2)}$, and the bounded parameter velocity bounds the remaining distance by $C|\log(a(T/2)/a(T))|$. The ratio tends to two. Since $\mathcal{D}$ decreases along the actual flow, integration gives $\mathcal{D}(H(T)) - \mathcal{D}(K) \leq C/T$. Adding the nonnegative comparison gap, at most $Ca(T)$, proves (105). □

The mean-curvature identity (14) makes $\|\Phi(T)\|_2^2$ nonincreasing. Therefore

$$\frac{T}{2}\|\Phi(T)\|_2^2 \le \int_{T/2}^{T} \|\Phi(t)\|_2^2 \,\mathrm{d}t \le \tfrac{1}{2}\{\mathcal{D}(H(T/2)) - \mathcal{D}_{\inf}\} \le C/T.$$

The unit-time scalar heat L^1-to-L^∞ estimate applied to $|\Phi|^2$ gives $\|\Phi(T)\|_\infty^2 \le C\|\Phi(T-1)\|_2^2$. This proves the first estimate of (99). The exact integrated inclusion identity then gives

$$\mathcal{E}(T) = \int_X \langle \Phi_H \iota, \iota\rangle \,\mathrm{d}V \le \|\Phi_H\|_\infty Z(T) \le C/T^2.$$

The lower bound follows from (37) and (88). This argument estimates the whole inclusion derivative and does not differentiate the mass equivalent.

9.2. **Smooth convergence on the intrinsic scale.** Fix p, a sequence $T_n \to \infty$, and a finite $R > 0$. Put $r_n = r_p(T_n)$ and use the rescaled cylinder $B_{4R}(0) \times [-4, 4]$, corresponding to

$$z = r_n y, \qquad T = T_n + r_n^2 \tau.$$

For large n it lies at positive times in the original coordinate ball. The rescaled base forms $\omega_n = r_n^{-2}\delta_n^*\omega$, where $\delta_n(y) = r_n y$, converge smoothly to the Euclidean form. Choose the time-independent scalar reference matrix

$$H_{0,n}(y) = \sqrt{h_L(r_n y) h_M(r_n y)}\, I.$$

It has the exact rescaled determinant, uniformly bounded smooth coefficients and inverse, and zero trace-free mean curvature.

Set $G_n(y,\tau) = H(T_n + r_n^2\tau, r_n y)$. These matrices converge uniformly on the cylinder to $H_{\mathrm{loc}}(y)$. Indeed $a' = O(a^2)$ implies $a(T_n + r_n^2\tau)/a(T_n) \to 1$ uniformly, $\alpha(T_n)/a(T_n) \to 1$, and the global metric comparison (95) tends to equality throughout the indicated times. The explicit reference metrics have the stated limit. In particular G_n and $H_{0,n}$ are uniformly comparable, and the relative logarithm is trace-free. The rescaled projective mean curvature is $r_n^2 \Phi_{H(T_n + r_n^2\tau)}$. It is uniformly bounded by (99); the original maximum-principle bound would already suffice here.

Apply the unit-radius $k = 0$ interior estimate of [JW18, Appendix C, Theorem C.1 and Proposition C.2] on nested balls at each time. Its inputs are bounded base and reference geometry, a bounded trace-free metric logarithm, and bounded projective mean curvature. All hold uniformly in these rescaled charts. The conclusion bounds the logarithm in $W^{2,p}$ on smaller balls for every finite p, hence in $C^{1,\beta}$ for $p > 4$. A finite covering gives a uniform spatial gradient bound on B_{3R}. We use the unit-radius conclusion after rescaling; metric comparability alone is not the hypothesis of this estimate.

We give the remaining parabolic step explicitly. In the fixed holomorphic frames the rescaled metric equation has a uniformly parabolic scalar principal part on matrix entries, with coefficients determined by ω_n. The remaining terms are smooth quadratic expressions in the first spatial derivatives, the positive matrix and its inverse, together with the smooth zeroth-order term $2\kappa r_n^2 G_n$. The spatial gradient bound makes these terms uniformly bounded. Interior parabolic $W_p^{2,1}$ estimates, with $p > 6$, give joint $C^{1+\beta,(1+\beta)/2}$ bounds on a smaller cylinder. The right side is then Hölder, and parabolic Schauder estimates give $C^{2+\beta,1+\beta/2}$. Differentiating the smooth equation and repeating on nested cylinders bounds every fixed parabolic derivative. At each step there is positive separation from the spatial and temporal boundaries. The reference matrix is independent of τ, so it contributes no reference velocity. This is the local bootstrap in [XYY26, proof of Theorem 2.5, (2.53)–(2.59)], with its prescribed-tensor term replaced here by the displayed smooth zeroth-order term.

We have proved smooth precompactness on smaller cylinders. Every subsequential smooth limit is the already determined uniform limit H_{loc}. A subsequence contradiction gives full smooth convergence at $\tau = 0$ on B_R. Since R and the sequence T_n were arbitrary, this proves the smooth convergence in Theorem 9.1. No rate for the higher derivatives is needed or asserted.

9.3. **Defect energy and the two radii.** Chern curvature depends smoothly on the metric, its inverse and its first two derivatives. The smooth inner convergence therefore implies convergence of curvature in the rescaled holomorphic frames and fibre metrics. Four-dimensional curvature energy is unchanged by the constant spatial dilation and corresponding base rescaling. By (19),

$$\int_{B_R} |F_{H_{\mathrm{loc}}}|^2 \, \mathrm{d}V_{\mathrm{E}} = 48\pi^2 \int_0^{R^2} \frac{s}{(1+s)^4} \, \mathrm{d}s = 8\pi^2 \mathfrak{F}(R).$$

This proves (100).

For completeness these local limits account for the full defect charge. An integrable unitary connection on a Kähler surface has self-dual part $(\Lambda F)\omega/2$. With our real norm, the self-dual decomposition gives

$$\int_X |F_H|^2 \, \mathrm{d}V = \int_X \mathrm{tr}(F_H \wedge F_H) + \int_X |\Lambda F_H|^2 \, \mathrm{d}V. \tag{107}$$

The second term is $2\kappa^2 v_0 + \|\Phi_H\|_2^2$. If $N = \#\Sigma$, the exact sequence implies

$$\mathrm{ch}_2(V) = \mathrm{ch}_2(L \oplus M) - N[\mathrm{pt}].$$

Since $\mathrm{ch}_2 = -\mathrm{tr}(F \wedge F)/(8\pi^2)$, the topological first terms for V and $L \oplus M$ differ by $8\pi^2 N$. Thus

$$\lim_{T\to\infty} \int_X |F_{H(T)}|^2 \, \mathrm{d}V = \int_X |F_{D_*}|^2 \, \mathrm{d}V + 8\pi^2 N. \tag{108}$$

This also checks the sign: the anti-self-dual local model contributes positive energy $8\pi^2$.

Fix disjoint radius-r coordinate balls and a fixed $R > 0$. The corrected connection convergence in Section 2 gives convergence of the gauge-invariant curvature energy on their fixed exterior. Subtracting that exterior energy and the core limits from (108) therefore gives

$$\begin{aligned} \lim_{T\to\infty} \sum_p \int_{B_r(p)\setminus B_{Rr_p(T)}(p)} |F_{H(T)}|^2 \, \mathrm{d}V &= 8\pi^2 N\{1 - \mathfrak{F}(R)\} \\ &\quad + \sum_p \int_{B_r(p)} |F_{D_*}|^2 \, \mathrm{d}V. \end{aligned} \tag{109}$$

Let $r \downarrow 0$ and then $R \uparrow \infty$. The right side tends to zero, proving tightness in this ordered-limit sense. Nonnegativity gives the same conclusion at each point. We assert no quantitative tail for a simultaneously varying R.

Finally $\mathfrak{F}$ is continuous and strictly increasing on $R > 0$, and $\mathfrak{F}(1) = 1/2$. For any fixed $0 < \delta < 1$, the actual energies inside $(1-\delta)r_p(T)$ and $(1+\delta)r_p(T)$ are eventually below and above $4\pi^2$, respectively. Energy in a ball is continuous and nondecreasing in its radius, so the first radius reaching that value lies between the two. Letting $\delta \downarrow 0$ proves (101) and completes the proof of Theorem 9.1. By Corollary 8.2, the first half-height inclusion radius has the same leading value, uniformly in direction. The energy threshold here is half the defect quantum, not half the total curvature energy of the compact bundle.

10. Consequences and examples

10.1. **Effective dissipation and logarithmic growth.** For the fixed inclusion define the effective cubic dissipation coefficient

$$A(T) = \frac{-Z'(T)}{Z(T)^3} = \frac{2\mathcal{E}(T)}{Z(T)^3}.$$

The adjective refers only to this scalar identity; it does not assert that Z solves an autonomous scalar equation.

Corollary 10.1. *Under the reduced, distinct-factor hypotheses,*

$$cT \le A(T) \le CT, \qquad \liminf_{T\to\infty} \frac{A(T)}{T} \ge C_*^2, \qquad \lim_{T\to\infty} \frac{2}{T^2}\int_0^T A(t)\,\mathrm{d}t = C_*^2. \tag{110}$$

On every compact subset of U,

$$\log u(T) + \log T \longrightarrow \log \frac{v_0}{4\pi^2\sigma} \tag{111}$$

uniformly. In particular $\log Z(T)/\log T \to -1$.

Proof. The bounds follow from $\mathcal{E}(T) \asymp T^{-2}$ and $Z(T) \sim C_*^{-1}T^{-1}$. For the lower limit, write

$$\frac{A(T)}{T} = \frac{-Z'(T)}{Z(T)^2}\,\frac{1}{TZ(T)}$$

and apply (37). The exact identity

$$Z(T)^{-2} = Z(0)^{-2} + 2\int_0^T A(t)\,\mathrm{d}t$$

and the mass equivalent prove the averaged limit. Finally $u/Z \to 1/v_0$ uniformly on each compact subset of U; its limit is positive. Combining this with (88) and taking logarithms proves (111). □

Neither (110) nor the mass equivalent asserts a pointwise equivalent for $\mathcal{E}$ or A. The mass equivalent excludes a nonconstant leading logarithmic multiplier in this class, but does not determine subleading terms.

There is a precise comparison with the two-weight description in [Wue26a, Section 3]. In the category of equal-slope semistable sheaves modulo sheaves supported at finitely many points, the extension has subobject chain $0 < L < V$. Its rank-polarized weight problem minimizes

$$w_L^2 + w_M^2 \quad \text{subject to} \quad w_M - w_L \ge 1,$$

giving $(-1/2, 1/2)$, with no further iterated weights. These are weights for squared metric norms. The proposed line metric $T^{-1/2}h_L$ consequently has the same two-weight normalization as the finite model in [HKKP23, Example 5.12]. Formula (111) gives the actual exponent -1 for the present singular class. Thus the difference from $-\frac{1}{2}\log T$ is $-\frac{1}{2}\log T + O(1)$, not merely an unspecified unbounded negative remainder. The local ideal, its Jacobian determinant in the fixed metrics, and the matching through the compact extension supply information which that polarized chain does not record.

10.2. **A toric tangent bundle.** We spell out the specialization to the tangent-bundle example of [Wue26a, Section 2]. Let X have fan with cyclic rays

$$(1,0), \quad (1,1), \quad (0,1), \quad (-1,0), \quad (1,-1).$$

Consecutive determinants are one. Removing $(1,1)$ gives the first Hirzebruch surface, so X is its blowup at the corresponding fixed point. With pullback section and fibre classes S, f and exceptional class E,

$$S^2 = -1, \quad Sf = 1, \quad f^2 = 0, \quad E^2 = -1, \quad SE = fE = 0.$$

Choose a Kähler form in $4S + 7f - 2E$, and put

$$V = T_X, \qquad L = \mathcal{O}_X(2S + f - E), \qquad M = \mathcal{O}_X(2f).$$

The torus direction $(1,0)$ gives a saturated tangent line. Its divisorial class is the sum of the two opposite-ray divisors, namely $2S + f - E$. On the chart of the cone $((1,1),(0,1))$, its inclusion and quotient have the presentation

$$\iota = (z_1, -z_2)^{\mathsf{t}}, \qquad q = (z_2, z_1).$$

All other maximal cones contain one of the two opposite rays; saturation there removes the divisorial zero and leaves a nonvanishing generator. Thus the only zero is the reduced fixed point p, and

$$0 \longrightarrow L \longrightarrow T_X \longrightarrow \mathcal{I}_p M \longrightarrow 0.$$

The intersection pairing gives

$$\deg_\omega L = \deg_\omega M = 8, \qquad v_0 = 18, \qquad \kappa = 8\pi/9, \qquad c_2(T_X) - c_2(L \oplus M) = 1.$$

In particular the two factors are distinct and all hypotheses of the positive theorem hold. Writing $\lambda = \lambda_p$ in the fixed normalization,

$$Z(T) \sim \frac{81\lambda}{\pi^2 T}, \qquad r_p(T) \sim \frac{3}{\sqrt{2}\,\pi}\, T^{-1/2}. \tag{112}$$

These follow by substituting $\sigma = \lambda^{-1}$. The volume factor is evaluated here; the geometric constant λ has not been numerically computed. The radius formula is in base-orthonormal coordinates. The stationary family obtained by changing the polarization in [Wue26a, Section 7] is a different family; its parameter is not flow time.

10.3. **Nonreduced points and repeated factors.** For arbitrary finite W, Theorem 4.2 gives $Z = O(T^{-1})$, and the presence of any nonreduced point gives $TZ \to 0$. The proof supplies neither a positive inner model nor a sharp nonreduced exponent. The simple-zero normalization in Lemma 2.2 is unavailable there. An extension to that setting must retain the full ideal germ, not just its length.

When $L \simeq M$, the real small spectral space has dimension three by (40). The projected correction and its parameter bounds retain those directions. The scalar equation (84), and hence our positive actual-flow comparison, are not proved for that case. In particular, one cannot remove two directions from the projection to infer the same scalar law.

Both exclusions occur in a concrete example. On a projective K3 surface take a length-two curvilinear scheme with local ideal (z_1^2, z_2). There is a locally free extension

$$0 \longrightarrow \mathcal{O}_X \longrightarrow V \longrightarrow \mathcal{I}_W \longrightarrow 0.$$

Indeed $K_X = \mathcal{O}_X$, and the Cayley–Bacharach condition for the Serre construction holds: a constant vanishing on a length-one subscheme vanishes on all of W. Apply [HL97, Theorem 5.1.1]; the local complete intersection condition is immediate from the displayed regular

sequence. The determinant is trivial, $c_2(V) = 2$, and both factors have degree zero for every Kähler form. The general nonreduced conclusion applies, but the reduced-point profile and radius theorem do not.

Finally, the positive result concerns a fixed compact surface and a fixed holomorphic extension. It does not classify singular graded sheaves of higher rank, varying holomorphic defects, or singular fibres in a degenerating threefold family. The centers here are fixed by W; their leading radii are linked through one amplitude and the constants λ_p. These are features of the stated problem, not a classification of all possible concentration parameters.

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